

**Unipotent algebraic groups. Extensions between
unipotent groups and groups of multiplicative type**
SGA 3, Exposé XVII

English draft

M. Raynaud

About this draft

This is an unofficial English translation of Exposé XVII, “Unipotent algebraic groups. Extensions between unipotent groups and groups of multiplicative type” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Expo17.pdf` (50 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé XVII. Unipotent algebraic groups. Extensions between unipotent groups and groups of multiplicative type

by M. Raynaud*

0. Some notation

In the present chapter, we shall mainly have to consider algebraic groups defined over a field k . The number $p \geq 0$ will always denote the characteristic of k , $\mathbf{F}_p$ the prime field with p elements if $p > 0$, $\bar{k}$ an algebraically closed extension of k , q a prime number distinct from p .

For every S -prescheme, $(\mathbf{G}_a)_S$ (resp. $(\mathbf{G}_m)_S$) denotes the additive group (resp. the multiplicative group) over S (cf. Exp. I 4.3). For every integer $n > 0$, $(\mu_n)_S$ (resp. $(\mathbf{Z}/n\mathbf{Z})_S$) denotes the group of n th roots of unity (Exp. I 4.4.4) (resp. the constant group over S associated with the abstract group $\mathbf{Z}/n\mathbf{Z}$ (Exp. I 4.1)). The group $(\mathbf{Z}/n\mathbf{Z})_S$ is finite and étale over S ; the group $(\mu_n)_S$ is flat and finite over S , and is étale over S if and only if n is invertible on S (Exp. VIII 2.1).

If S is a prescheme of characteristic $p > 0$, for every integer $n > 0$ and every group S -prescheme G , we denote by $_{F^n}(G)$ the radicial group S -subprescheme of G equal to the kernel of the n th iterate of the Frobenius morphism relative to G (Exp. VIIA). In particular, if $G = (\mathbf{G}_a)_S$, we put $_{F^n}(G) = (\alpha_{p^n})_S$, which is a radicial S -group, flat and finite over S , representing the following functor: for every S -prescheme S' , $(\alpha_{p^n})_S(S')$ is the set of $x' \in \Gamma(S', \mathcal{O}_{S'})$ such that $x'^{p^n} = 0$.

The group $(\mu_{p^n})_S$, already defined, is canonically isomorphic to $_{F^n}(\mathbf{G}_m)_S$.

When there is no ambiguity about the base scheme S , we shall simply write $\mathbf{G}_a, \mathbf{G}_m, \alpha_{p^n}$, etc., instead of $(\mathbf{G}_a)_S, (\mathbf{G}_m)_S, (\alpha_{p^n})_S$, etc.

If G is a commutative group S -prescheme, for every integer $n > 0$, $_nG$ is the group subprescheme of G equal to the kernel of raising to the n th power in G .

For the reader's convenience, we have collected in an appendix some properties of algebraic groups proved in Exp. VI and VII, as well as elementary properties of Hochschild cohomology, which will be useful to us in this chapter.

1. Definition of unipotent algebraic groups

Definition 1.1. An algebraic group G defined over an algebraically closed field k is said to be *unipotent* if G admits a composition series whose successive quotients are isomorphic to algebraic subgroups of $(\mathbf{G}_a)_k$.

Proposition 1.2. *Let k be an algebraically closed field, K an algebraically closed extension of k , G an algebraic group defined over k . Then, for G to be unipotent, it is necessary and sufficient that G_K be unipotent.*

The necessity of the condition is clear, since a composition series gives a composition series by extension of the base.

* Cf. the note on page 1 of Exposé XV.

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The proof of sufficiency is standard: K is the inductive limit of its finitely generated k -subalgebras. By Exp. VIB § 10, one can find a finitely generated k -subalgebra A of K , a composition series G_i of G_S ($S = \text{Spec } A$), and immersions $u_i : H_i = G_i/G_{i+1} \rightarrow (\mathbf{G}_a)_S$. To prove that G is unipotent, it then suffices to make a k -extension of the base, $A \rightarrow k$, which is possible since $\text{Hom}_{k\text{-alg}}(A, k)$ is nonempty, A being a nonzero finitely generated k -algebra over the algebraically closed field k .

Definition 1.3. Let G be an algebraic group defined over a field k . We shall say that G is *unipotent* if there is an algebraically closed extension $\bar{k}$ of k such that $G_{\bar{k}}$ is unipotent (Definition 1.1).

By 1.2, the property is independent of the chosen algebraically closed extension $\bar{k}$.

Definition 1.4. Let G be an algebraic group defined over a field k , and H an algebraic group defined over an extension k' of k . We shall say that H is a *form* of G over k' if the algebraic groups $G_{k'}$ and $H_{k'}$ become isomorphic over $\bar{k}'$ (as above, one sees that the property does not depend on the choice of the algebraically closed extension $\bar{k}'$ of k'). We shall also say that H is a “twisted” group G .

We are now in a position to describe the algebraic subgroups of $\mathbf{G}_a$.

Proposition 1.5. *Let k be a field of characteristic $p \geq 0$. Then an algebraic subgroup H of $(\mathbf{G}_a)_k$ is of one of the following types:*

- (i) $H = 0$.
- (ii) $H = \mathbf{G}_a$.
- (iii) (If $p > 0$) H is an extension of a twisted constant group $(\mathbf{Z}/p\mathbf{Z})^r$ by a radicial group α_{p^n} (r and n integers ≥ 0 , $r + n > 0$). If, moreover, k is perfect, this extension is necessarily trivial.

Proof. If H is of dimension 1, it is clear that $H = \mathbf{G}_a$. Otherwise H is of dimension 0 and consequently is an extension⁽¹⁾ of an étale group H'' by its neutral component H' , which is a radicial group. To describe H''_k , it suffices to know the abstract group $H''(\bar{k})$, isomorphic to $H(\bar{k})$. Now the latter is a finite subgroup of $\mathbf{G}_a(\bar{k})$, hence is zero if $p = 0$ and is of the form $(\mathbf{Z}/p\mathbf{Z})^r$ otherwise, since it is annihilated by p . The group H' is closed in $\mathbf{G}_a$ and is defined by a single equation (the ring $k[T]$ of $\mathbf{G}_a$ is principal), which over $\bar{k}$ has 0 as its only root; hence this equation is of the form $T^n = 0$. Compatibility with the group law implies that $(T + T')^n$ belongs to the ideal generated by T^n and T'^n in the ring $k[T, T']$; hence: if $p = 0$, one has $n = 1$ and H' is the unit group; if $p > 0$, one has $n = p^m$ and $H' = \alpha_{p^m}$.

The last assertion of 1.5 follows more generally from the lemma:

Lemma 1.6. *If k is a perfect field, every extension H of an étale algebraic group H'' by a radicial group H' is trivial. Moreover, there is a unique lifting of H'' into H , namely H_{red} .*

Indeed, since k is perfect, the reduced k -scheme H_{red} is an algebraic subgroup of H (Exp. VIA 0.2), geometrically reduced, hence smooth over k (Exp. VIA, 1.3.1), hence étale, since H is of dimension 0. To see that the canonical projection $H_{\text{red}} \rightarrow H''$ is an isomorphism, it suffices to see it after extension of the base field $k \rightarrow \bar{k}$, in which case it suffices to show that one has an isomorphism on the points with values in $\bar{k}$, which is quite clear. The last assertion follows from the fact that every lifting of H'' into H , being étale over k , is reduced, hence is necessarily contained in H_{red} .

Note that α_{p^n} is a multiple extension of groups isomorphic to α_p . One then deduces from 1.5 the corollary:

⁽¹⁾**N.D.E.** Reference to VIIA?

Corollary 1.7. *For an algebraic group G defined over an algebraically closed field k to be unipotent, it is necessary and sufficient that it have a composition series whose successive quotients are isomorphic to $\mathbf{G}_a$ if $p = 0$, and to one of the groups $\mathbf{G}_a, \mathbf{Z}/p\mathbf{Z}, \alpha_p$ if $p > 0$. (We shall call these groups the elementary unipotent groups.)*

2. First properties of unipotent groups

Proposition 2.1. *A unipotent algebraic group defined over a field k is affine over k .*

By (fpqc) descent of affine morphisms, it suffices to prove 2.1 when k is algebraically closed. In this case, G is, by definition, a multiple extension of affine algebraic groups, hence is affine, by Exp. VIB 9.2(viii) applied to affine morphisms.⁽²⁾

Proposition 2.2.

- i) *The property for an algebraic group of being unipotent is invariant under extension of the base field.*
- ii) *Every algebraic subgroup of a unipotent group is unipotent.*
- iii) *Every algebraic quotient group of a unipotent group is unipotent.*
- iv) *Every extension of a unipotent algebraic group by a unipotent algebraic group is likewise unipotent.*

Proof. i) follows immediately from 1.3 and 1.2. To establish the other properties, we can assume that the field k is algebraically closed. Then iv) is obvious from Definition 1.3.

Let therefore G be a unipotent algebraic group, G' an algebraic subgroup of G , G'' an algebraic quotient group of G , G_i ($i = 1, \dots, n$) a composition series of G such that $H_i = G_i/G_{i+1}$ is an elementary unipotent group (1.7).

To prove ii), consider the composition series of G' induced by that of G : $G'_i = G_i \cap G'$. The group G'_i/G'_{i+1} is identified with an algebraic subgroup of H_i , hence is isomorphic to a subgroup of $\mathbf{G}_a$, and consequently G' is unipotent.

To prove iii), consider the composition series of G'' that is the image of that of G : $G''_i = \text{image of } G_i \text{ in } G''$. The group G''_i/G''_{i+1} is then a quotient of H_i , and it suffices to prove the lemma:

Lemma 2.3. *If H is an elementary unipotent group (1.7) defined over a field k , every algebraic quotient group H'' of H is zero or is isomorphic to H over k .*

Proof. If $H = \mathbf{G}_a$ in characteristic 0, or if $H = \alpha_p$ or $\mathbf{Z}/p\mathbf{Z}$ ($p > 0$), it suffices to note that it follows from 1.5 that H has no algebraic subgroups other than 0 and H (observe that a nonzero algebraic subgroup of α_p is defined by a k -algebra of rank over k at least equal to p (1.5 iii), hence is equal to α_p).

Let now $H = \mathbf{G}_a$ ($p > 0$), so that one has an exact sequence:

$$0 \longrightarrow N \longrightarrow \mathbf{G}_a \longrightarrow H'' \longrightarrow 0.$$

If $N = \mathbf{G}_a$, then $H'' = 0$. Otherwise, proceeding by induction on the length of a composition series of N , one can assume that $N = \alpha_p$, or that N is a form of $(\mathbf{Z}/p\mathbf{Z})^r$ (1.5).

a) If $N \simeq \alpha_p$, the proof of 1.5 iii) shows that N is necessarily the kernel of the Frobenius morphism F in $\mathbf{G}_a$, and one concludes with the aid of the exact sequence:

$$0 \longrightarrow \alpha_p \longrightarrow \mathbf{G}_a \xrightarrow{F} \mathbf{G}_a \longrightarrow 0.$$

b) If $N \simeq \mathbf{Z}/p\mathbf{Z}$, it is immediate that there is $a \in k^*$ such that N is the closed subscheme of $\mathbf{G}_a = \text{Spec } k[X]$ defined by the equation $X^p - aX = 0$. One then concludes with the aid of the exact sequence:

$$0 \longrightarrow N \longrightarrow \mathbf{G}_a \xrightarrow{\mathcal{P}} \mathbf{G}_a \longrightarrow 0,$$

⁽²⁾**N.D.E.** Check this reference.

where $\mathcal{P}$ is the Artin–Schreier morphism: $x \mapsto x^p - ax$.

c) If N is a form of $(\mathbf{Z}/p\mathbf{Z})^r$, there is a finite Galois extension k' of k trivializing N . By b) and an obvious induction on r , $\mathbf{G}_a/N$ is a form of $\mathbf{G}_a$ trivialized by k' . It then suffices to apply the following lemma:

Lemma 2.3 bis. *Let k be a field, G an algebraic k -group that is a form of $\mathbf{G}_a$, trivialized by a finite separable extension k' of k . Then G is isomorphic to $\mathbf{G}_a$.*

Indeed, the group of k' -automorphisms of the algebraic group $(\mathbf{G}_a)_{k'}$ is the group of nonzero homotheties $(k')^\times$ (*Bible*, Exp. 9, Lemma 1), and the Galois cohomology group $H^1(k'/k, \mathbf{G}_m)$ is zero (one can assume that k' is a Galois extension of k), by Hilbert’s Theorem 90. Lemma 2.3 bis then follows from the classification of k -forms of an algebraic group (cf. J.-P. Serre, *Cohomologie galoisienne*, Chap. III, 1.3).

This completes the proof of 2.2.

Proposition 2.4. *Let k be a field, M a k -group of multiplicative type (Exp. VIII) and of finite type, U a unipotent algebraic k -group. Then:*

- i) $\underline{\mathrm{Hom}}_{k\text{-gr}}(M, U) = \underline{e}$ (a fortiori $\mathrm{Hom}_{k\text{-gr}}(M, U) = e$).
- ii) $\mathrm{Hom}_{k\text{-gr}}(U, M) = e$.

To prove i), we must establish that for every prescheme S over k , $\mathrm{Hom}_{S\text{-gr}}(M_S, U_S) = e$. But this follows from the following lemma:

Lemma 2.5. *Let S be a prescheme, M an S -group of multiplicative type and of finite type over S , U a group S -prescheme, of finite presentation over S , with unipotent fibers; then $\mathrm{Hom}_{S\text{-gr}}(M, U) = e$.*

Indeed, under the hypotheses made, for a group S -morphism $u : M \rightarrow U$ to be the unit homomorphism, it suffices that the restriction of u to the fibers of M over the points of S be the zero morphism (Exp. IX 5.2). We are therefore reduced to the case where S is the spectrum of a field, which one can moreover assume to be algebraically closed. In view of Definition 1.1, one can restrict oneself to $U = \mathbf{G}_a$, in which case the property has already been proved (Exp. XII 4.4.1).

Let us now prove 2.4 ii). Let therefore $u : U \rightarrow M$ be a group k -morphism. The image $u(U)$ is representable by an algebraic subgroup U'' of M (Exp. VIB 5.4).⁽³⁾ The group U'' is unipotent as a quotient of a unipotent group (2.2 iii)) and is of multiplicative type as a subgroup of a group of multiplicative type (cf. *Bible*, Exp. 4, Th. 2 cor. 1, or Exp. IX 6.8), hence U'' is the unit group by 2.4 i).

Remark 2.6. Keeping the notation of 2.4, it is no longer true in general that the functor $\underline{\mathrm{Hom}}_{k\text{-gr}}(U, M)$ is equal to $\underline{e}$. Thus, take a prescheme S such that $\Gamma(S, \mathcal{O}_S)$ contains a nonzero element ε such that $\varepsilon^2 = 0$ (for example, the spectrum of the algebra of dual numbers of a ring A). For every S' over S , the map $u \mapsto 1 + \varepsilon_{S'} u$ defines a homomorphism, functorial in S' , from the additive group $\Gamma(S', \mathcal{O}_{S'})$ to the multiplicative group $\Gamma(S', \mathcal{O}_{S'}^\times)$, hence defines a group S -morphism $(\mathbf{G}_a)_S \rightarrow (\mathbf{G}_m)_S$, and since $\varepsilon \neq 0$, this morphism is nonzero.

Recall (Exp. VIIA § 3) that when G is a commutative group S -prescheme, finite and flat over S , one has Cartier duality, and G is reflexive in the sense of Exp. VIII § 1. More precisely, the functor $G \mapsto \underline{\mathrm{Hom}}_{S\text{-gr}}(G, \mathbf{G}_m)$ is representable by a commutative group S -prescheme $D(G)$, finite and flat over S , and the canonical morphism $G \rightarrow D(D(G))$ is an isomorphism. In particular, one obtains:

- (i) $D(\mu_n) \simeq \mathbf{Z}/n\mathbf{Z}$, and consequently $D(\mathbf{Z}/n\mathbf{Z}) \simeq \mu_n$.
- (ii) If S is of characteristic $p > 0$, $D(\alpha_p) \simeq \alpha_p$.

⁽³⁾**N.D.E.** Check this reference.

3. Unipotent groups acting on a vector space

Recall (Exp. I, 4.6.1) that if S is a prescheme and M a sheaf of $\mathcal{O}_S$ -modules, one denotes by $W(M)$ the S -functor $((\text{Sch})/S)^\circ \rightarrow (\text{Ens})$ defined by the condition $W(M)(S') = \Gamma(S', M \otimes_{\mathcal{O}_{S'}} \mathcal{O}_{S'})$ for every S -prescheme S' .

Moreover, recall that if an S -group acts on an S -functor $\mathbf{V}$, one defines the S -functor $\mathbf{V}^G$ of invariants of $\mathbf{V}$ under G to be the subfunctor of $\mathbf{V}$ whose set of points with values in an S' over S is the set of $x \in \mathbf{V}(S')$ such that $x_{S''}$ is fixed under $G(S'')$ for every S'' over S' .

This being so, one has the following lemma:

Lemma 3.1. *Let S be a prescheme, G a group S -prescheme, affine over S , defined by the quasi-coherent $\mathcal{O}_S$ -algebra A . Suppose that G acts on a quasi-coherent sheaf of $\mathcal{O}_S$ -modules M , and let $\mu : M \rightarrow A \otimes_{\mathcal{O}_S} M$ be the comorphism defining the action of G on M (Exp. I 4.7.2), and $\nu : M \rightarrow A \otimes_{\mathcal{O}_S} M$ the morphism $x \mapsto \mu(x) - 1 \otimes x$. Then:*

- i) $\underline{M}^G(S) = \Gamma \text{Ker}(\nu)$.
- ii) *If S is the spectrum of a field k , $\underline{M}^G$ is of the form $W(N)$, where N is a vector subspace of M .*
- iii) *If S is the spectrum of a field k , every element x of M is contained in a vector subspace of M , finite-dimensional over k , stable under the action of G .*

Proof. iii) is included as a reminder and has already been proved in Exp. VIB 11.2.

i) It is clear that $\underline{M}^G(S)$ contains $\Gamma \text{Ker}(\nu)$. To establish the converse, one can assume that S is affine with ring B . Let $m \in \underline{M}^G(S)$. Then, for every B -algebra B' and every element u of $\text{Hom}_{B\text{-alg}}(A, B')$ (u corresponds to an element of $G(\text{Spec } B')$), one has:

$$(u \otimes 1_M)\nu(m) = 0 \quad \text{in } B' \otimes_B M.$$

Take, in particular, $B' = A$ and u the identity of A ; one does indeed find $\nu(m) = 0$.

ii) Let N be the kernel of ν , equal to $\underline{M}^G(k)$ by i). Every k -prescheme S is flat over k , hence one has:

$$\underline{M}^G(S) = \Gamma \text{Ker}(\nu \otimes_k S) = \Gamma(N \otimes_k S).$$

Thus $\underline{M}^G$ is isomorphic to the functor $W(N)$.

Proposition 3.2. *Let G be a unipotent algebraic group defined over a field k , acting on a k -vector space V . Then, if $V \neq 0$, one has $V^G \neq 0$.*

In view of 3.1 ii) and iii), one can assume that k is algebraically closed and V is finite-dimensional over k .

Let $0 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 0$ be an exact sequence of algebraic groups, and suppose that G acts on a sheaf $\mathbf{V}$ (for the fpqc topology). One has, of course, $\mathbf{V}^G \subset \mathbf{V}^{G'}$, and the quotient presheaf G/G' acts naturally on $\mathbf{V}^{G'}$. But $\mathbf{V}^{G'}$ is a sheaf (as the kernel of the well-known pair of morphisms $\mathbf{V} \rightrightarrows \underline{\text{Hom}}(G', \mathbf{V})$); consequently, the sheaf associated with G/G' , that is, G'' , acts on $\mathbf{V}^{G'}$, and it is immediate to check that $(\mathbf{V}^{G'})^{G''} = \mathbf{V}^G = (\mathbf{V}^{G'})^{G/G'}$.

This remark allows us, in order to prove 3.2, to reduce to the case where G is an elementary unipotent group (1.7).

a) $G = \mathbf{G}_a$, $p = 0$. It follows from (BIBLE 4 prop. 4) that a morphism from $\mathbf{G}_a$ to the linear group $\text{GL}(V)$ is given by an exponential map:

$$T \mapsto \sum_{q=0}^{\infty} T^q \frac{n^q}{q!},$$

where n is a nilpotent endomorphism of V . But then $V \neq 0 \Rightarrow \text{Ker } n \neq 0$, and it is clear that every vector of V annihilated by n is left fixed by G .

Suppose now that $p > 0$.

b) $G = \alpha_p$. Since the group α_p is a radicial group of height 1 (Exp. VIIA § 7), giving a representation of α_p in V amounts to giving a representation of the Lie p -algebra α_p in $\mathfrak{gl}(V)$

(App. II 2.2), that is, here, to giving an element X of $\text{End}(V)$ such that $X^p = 0$ (App. II 2.1). But then $V \neq 0 \Rightarrow W = \text{Ker}(X) \neq 0$, and, again by (App. II 2.2), one has $W = V^{\alpha_p}$.

c) $G = \mathbf{Z}/p\mathbf{Z}$. A representation of G in V is equivalent to giving an element x of $\text{Aut}(V)$ such that $x^p = 1$, i.e. $(1 - x)^p = 0$, hence x is of the form $1 + n$, with n nilpotent, and $W = \text{Ker } n$ is left fixed by x .

d) $G = \mathbf{G}_a$. Let G_i , $i \in I$, be the increasing filtered family of étale algebraic subgroups of $\mathbf{G}_a$, hence isomorphic to $(\mathbf{Z}/p\mathbf{Z})^{r_i}$ (Prop. 1.5), and let $V_i = V^{G_i}$. Since V is finite-dimensional and nonzero, and V_i is nonzero by c), the decreasing filtered family of the V_i is stationary, and $W = \bigcap_{i \in I} V_i \neq 0$. Now one has the lemma:

Lemma 3.3. *The family of étale subgroups of $(\mathbf{G}_a)_S$ (S a prescheme of characteristic $p > 0$) is schematically dense in G (Exp. IX 4.1).*

By Exp. IX 4.4, it suffices to prove the lemma when S is the spectrum of the prime field $\mathbf{F}_p$. In this case, it suffices to consider the family of étale subgroups G_n ($n \geq 1$) with equation $X^{p^n} - X = 0$, which is schematically dense in $(\mathbf{G}_a)_{\mathbf{F}_p}$, since it contains every closed point.

This being so, let us return to the proof of 3.2 d). If $w \in W$, its stabilizer in G is an algebraic subgroup of $\mathbf{G}_a$ containing G_i for every $i \in I$, hence is equal to $\mathbf{G}_a$ (3.3), and consequently $W = V^G$.

One immediately deduces from 3.2 the

Corollary 3.4. *Let k be a field, G a unipotent algebraic k -group acting on a finite-dimensional k -vector space V . Then V has a sequence of vector subspaces V_i , defined over k , stable under G ,*

$$0 = V_0 \subset V_1 \subset \cdots \subset V_n = V,$$

such that G acts trivially on V_{i+1}/V_i . One can moreover assume that V_{i+1}/V_i is of dimension 1.

We shall now summarize and complete the already proved properties of unipotent groups in the following theorem:

Theorem 3.5. *Let G be an algebraic group defined over a field k . The following properties are equivalent:*

- i) G is unipotent.
- ii) G has a composition series, defined over k , whose successive quotients are isomorphic to $\mathbf{G}_a$ if $p = 0$ (resp. to α_p , $\mathbf{G}_a$, or a twisted $(\mathbf{Z}/p\mathbf{Z})^r$ (1.4) if $p > 0$).
- iii) As in ii), but one moreover assumes that the composition series is central.
- iv) G has a characteristic composition series (Exp. VIB) defined over k , whose successive quotients are isomorphic to $(\mathbf{G}_a)^r$ if $p = 0$ (resp. to $(\alpha_p)^r$, a twisted $(\mathbf{G}_a)^s$ or a twisted $(\mathbf{Z}/p\mathbf{Z})^t$, taken in this order, if $p > 0$).
- v) G is isomorphic to an algebraic subgroup of the group $\text{Trigstr}(n)_k$ of strict upper triangular matrices of the linear group $\text{GL}(n)_k$, for a suitable integer $n \geq 0$.
- vi) G is affine, and for every linear representation of G in a nonzero finite-dimensional k -vector space V , one has $V^G \neq 0$.

Proof.

i) $\Rightarrow$ vi) by 2.1 and 3.2.

vi) $\Rightarrow$ v). Since the algebraic group G is affine, G is an algebraic subgroup of a suitable linear group $\text{GL}(V)$ (Exp. VIB 11.3). Apply 3.4 to the representation of G in V defined by this embedding; one finds v).

v) $\Rightarrow$ iii). One knows that the algebraic group $\text{Trigstr}(n)$ has a central composition series with successive quotients isomorphic to $\mathbf{G}_a$. The composition series induced on G gives property iii), in view of 1.5.

iii) $\Rightarrow$ ii) $\Rightarrow$ i), and iv) $\Rightarrow$ i), are clear.

We shall prove $i) \Rightarrow iv)$ in a moment, but let us already note a few consequences of what has been proved.

Definition 3.6. We shall say that a Lie p -algebra $\mathfrak{g}$ ($p > 0$) (cf. Exp. VIIA § 5) is *unipotent* if the map $x \mapsto x^{(p)}$ is nilpotent, that is, if for every $x \in \mathfrak{g}$ there is an integer $n > 0$ such that $x^{(p^n)} = 0$.

Corollary 3.7. *A unipotent algebraic group G is nilpotent (Exp. VIB § 8); its Lie algebra $\mathfrak{g}$ is nilpotent (Bourbaki, Groupes et algèbres de Lie, Chap. 1 § 4) and is isomorphic to a Lie algebra of nilpotent endomorphisms of a finite-dimensional vector space. In characteristic $p > 0$, $\mathfrak{g}$ is a unipotent Lie p -algebra (3.6).*

Since $i) \Rightarrow v)$, it suffices to prove 3.7 when $G = \text{Trigstr}(n)$. We have already used the fact that $\text{Trigstr}(n)$ is a nilpotent algebraic group. Moreover, the Lie algebra $\mathfrak{h}$ of $\text{Trigstr}(n)$ consists of the upper triangular endomorphisms of V that have zeros on the main diagonal. They are therefore nilpotent, and consequently $\mathfrak{h}$ is nilpotent (Bourbaki, *loc. cit.*, Chap. 1 § 4, cor. 3). If $p > 0$, since the p th power in the Lie p -algebra $\mathfrak{gl}(V) = \text{End}(V)$ coincides with the p th power of endomorphisms of V (Exp. VIIA 6.4.4), one sees that $\mathfrak{h}$ is unipotent.

Corollary 3.8. *Let k be an algebraically closed field, and let G be an algebraic group smooth and affine over k . Then the following properties are equivalent:*

- i) G is unipotent.
- ii) $G(k)$ consists of unipotent elements (BIBLE 4 prop. 4 cor. 1), that is, G is unipotent in the sense of BIBLE.

$i) \Rightarrow ii)$, since G is isomorphic to an algebraic subgroup of a group $\text{Trigstr}(n)$ by 3.2 $i) \Rightarrow v)$. $ii) \Rightarrow i)$. Let therefore G be a unipotent algebraic group in the sense of BIBLE, and denote its neutral component by G^0 . Since the maximal tori of G^0 consist of unipotent elements, they are trivial, hence G^0 is equal to its Cartan subgroups. Consequently, G^0 is solvable (BIBLE 6 Th. 6), hence triangularizable (BIBLE 6, Th. 1). Thus G^0 is an algebraic subgroup of a group $\text{Trigstr}(n)$, and is therefore unipotent in the sense of this exposé.

The group $(G/G^0)(k)$ is a finite group consisting of unipotent elements; it is therefore zero if $p = 0$ and equal to a finite p -group if $p > 0$ (BIBLE 4 prop. 4). But then G/G^0 is unipotent in the sense of this exposé, as a multiple extension of groups isomorphic to $\mathbf{Z}/p\mathbf{Z}$. This proves that G is unipotent.

End of the proof of 3.5. Let us prove $i) \Rightarrow iv)$.

a) $p > 0$. Consider the increasing sequence of algebraic subgroups of G :

$$\{e\} \subset F(G) \subset F^2(G) \subset \cdots \subset F^n(G) \subset G^0 \subset G.$$

One thus obtains a characteristic composition series of G (App. II 1), and, for n sufficiently large, $G/F^n(G)$ is smooth (App. II 3.1), so that the successive quotients are, in order:

- (1) radicial groups of height 1,
- (2) a smooth connected group,
- (3) an étale group.

To prove $i) \Rightarrow iv)$, it therefore suffices for us to prove:

Lemma 3.9. *Let G be a unipotent algebraic group defined over a field k of characteristic $p > 0$. Then G has a characteristic composition series, defined over k , whose successive quotients are isomorphic to:*

- i) $(\alpha_p)^r$ if G is radicial.
- ii) a twisted $(\mathbf{G}_a)^r$ if G is smooth and connected.
- iii) a twisted $(\mathbf{Z}/p\mathbf{Z})^r$ if G is étale.

Proof. i) The group G is radicial. Filtering G by the $F^n(G)$, one reduces to the case where G is radicial of height 1. Since G is nilpotent (3.5 $i) \Rightarrow iii)$), and since the center of an

algebraic group is representable (Exp. VIII 6.5 e)), one can consider the ascending central series of G , which is obviously characteristic in G , reducing us to the case where G is moreover commutative.

Let $\mathfrak{g} = \text{Lie}(G)$. The p th-power morphism π is therefore additive in $\mathfrak{g}$ (Exp. VIIA); we shall reduce to the case where it is zero. For every prescheme S over k , put $\mathfrak{g}_S = \mathfrak{g} \otimes_k \mathcal{O}_S$, and let $\mathfrak{h}_S$ be the subsheaf of abelian groups of $\mathfrak{g}_S$ that is the image of $\mathfrak{g}_S$ under π_S . Finally, let $\bar{\mathfrak{h}}_S$ be the subsheaf of $\mathcal{O}_S$ -modules of $\mathfrak{g}_S$ generated by $\mathfrak{h}_S$. It is clear that $\bar{\mathfrak{h}}_S = \bar{\mathfrak{h}}_k \otimes_k \mathcal{O}_S$, and that $\bar{\mathfrak{h}}_S^{(4)}$ is a characteristic Lie p -subalgebra of $\mathfrak{g}_S$ (that is, it is stable under the S -functor $\text{Aut}_{p\text{-Lie}}(\mathfrak{g})$). It then follows from App. II 2.2 that $\bar{\mathfrak{h}}_k$ is the Lie algebra of an algebraic subgroup H of G characteristic in G .

Moreover, in view of 3.7 and Lemma 3.9 bis below, if $G \neq \{e\}$, H is distinct from G , since $\bar{\mathfrak{h}}_k$ is distinct from $\mathfrak{g}$. Furthermore, if $G/H = G''$, one has $\text{Lie } G'' = \mathfrak{g}/\bar{\mathfrak{h}}_k$ (App. II 2.2), and consequently the p th power is zero in $\text{Lie } G''$. Proceeding by induction on $\dim \text{Lie } G$, one is therefore reduced to the case where $\text{Lie } G$ is a Lie p -algebra in which the p th power is zero. But then $\text{Lie } G''$ is isomorphic to $\text{Lie}(\alpha_p)^r$ for a suitable integer $r \geq 0$ (App. II 2.1), and consequently (App. II 2.2) G'' is isomorphic to $(\alpha_p)^r$. It remains to prove

Lemma 3.9 bis. *Let k be a field of characteristic $p > 0$, $\mathfrak{g}$ a commutative unipotent Lie p -algebra (3.6), finite-dimensional over k , and $\mathfrak{h}$ the Lie p -subalgebra of $\mathfrak{g}$ generated by the image of the p th power in $\mathfrak{g}$. Then, if $\mathfrak{g} \neq 0$, one has $\mathfrak{h} \neq \mathfrak{g}$.*

Indeed, since $\mathfrak{g}$ is commutative, $\mathfrak{h}$ is simply the k -vector subspace of $\mathfrak{g}$ generated by the $X^{(p)}$ ($X \in \mathfrak{g}$). If $\mathfrak{g} \neq 0$ and is unipotent, there is $X \in \mathfrak{g}$, $X \neq 0$, such that $X^{(p)} = 0$. Let $X_1, \dots, X_n$ be a basis of a complement in $\mathfrak{g}$ to the line kX . The Lie algebra $\mathfrak{h}$ is then the k -vector subspace of $\mathfrak{g}$ generated by $X_1^{(p)}, \dots, X_n^{(p)}$, hence is of dimension at most $n = \dim \mathfrak{g} - 1$.

Proof of 3.9 ii). G is smooth and connected. In this case, the descending central series of G is representable by smooth connected characteristic algebraic subgroups G_i (Exp. VIB 8.3 and 7.4), and $G_i = 0$ for i sufficiently large, since G is nilpotent (3.5 i) $\Rightarrow$ iii)). It suffices to prove 3.9 for the groups G_i/G_{i+1} , reducing us to the case where G is moreover commutative. For every integer $n > 0$, let G_n be the algebraic subgroup of G that is the image of G under the morphism of raising to the p^n th power. The group G_n is therefore smooth, connected and characteristic, and it follows from Definition 1.1 of unipotent groups that $G_n = 0$ for n sufficiently large. Replacing G by G_n/G_{n+1} , one can moreover assume that G is annihilated by raising to the p th power. But then, by (J.-P. Serre, *Groupes algébriques et corps de classes*, chap. VII, prop. 11), G is a form of $(\mathbf{G}_a)^r$, for a suitable integer r . *Proof of 3.9 iii).* G is étale. Proceeding as in ii), one reduces to the case where G is commutative, then to the case where G is annihilated by p ; but then $G_{\bar{k}}$ is isomorphic to $(\mathbf{Z}/p\mathbf{Z})^r$.

Proof of 3.5 i) $\Rightarrow$ iv) in case b), $p = 0$. The group G is then smooth and connected, and, proceeding as in 3.9 ii), one reduces to the case where G is moreover commutative. One then has the following more precise result:

Lemma 3.9 ter. *Let k be a field of characteristic 0, G a commutative unipotent algebraic k -group, $\mathfrak{g} = \text{Lie } G$. Then there is a canonical isomorphism:*

$$\exp : W(\mathfrak{g}) \xrightarrow{\sim} G.$$

The morphism $\exp$ is the unique homomorphism $W(\mathfrak{g}) \rightarrow G$ inducing the identity on the Lie algebras.

Since G is unipotent, G can be realized as an algebraic subgroup of $\text{Trigstr}(n)$ for a suitable integer n (3.5 i) $\Rightarrow$ v)). The choice of such an embedding allows one to identify $\mathfrak{g}$

(4) **N.D.E.** One has changed $\mathfrak{h}$ to $\bar{\mathfrak{h}}$.

with a Lie subalgebra of $\mathfrak{gl}(n)$ consisting of nilpotent endomorphisms. Hence a k -morphism:

$$\exp : W(\mathfrak{g}) \longrightarrow \mathrm{GL}(n), \quad T \longmapsto \sum_{i \geq 0} \frac{T^i}{i!}.$$

Since G is commutative, the morphism $\exp$ is a homomorphism. Let G' be the algebraic group that is the image of $W(\mathfrak{g})$ under the morphism $\exp$. If one canonically identifies $\mathrm{Lie} W(\mathfrak{g})$ with $\mathfrak{g}$, the linear tangent map to $\exp$ is simply the injection $\mathfrak{g} \longrightarrow \mathfrak{gl}(n)$. Consequently, $\mathrm{Lie}(G \cap G') = \mathfrak{g} \cap \mathfrak{g} = \mathfrak{g}$. Since $G \cap G'$ is smooth (Exp. VIB 1.6.1), and G is connected (since it is a multiple extension of groups $\mathbf{G}_a$ (3.5 i) $\Rightarrow$ ii)), one necessarily has $G = G'$. The kernel $\mathrm{Ker}(\exp)$ is an étale unipotent group, hence is the unit group, and consequently $\exp$ is an isomorphism from $W(\mathfrak{g})$ onto G .

If $h : W(\mathfrak{g}) \longrightarrow G$ is another homomorphism such that $\mathrm{Lie}(h)$ is the identity map of $\mathfrak{g}$, the morphism $h - \exp$ is a homomorphism (G is commutative) whose linear tangent map is zero. Since k is of characteristic 0 and $W(\mathfrak{g})$ is connected, it again follows from Cartier's theorem that one necessarily has $h = \exp$.

Proposition 3.10. *Let G be an algebraic group defined over an algebraically closed field k . Then the following properties are equivalent:*

- i) Every morphism from a group of multiplicative type M to G is the zero morphism.
- i bis) G has no nonzero subgroups of multiplicative type.
- ii) a) if $p = 0$: $G(k)$ contains no points of finite order other than e ;
 b) if $p \neq 0$: for every prime number $q \neq p$, $x \in G(k)$ and $x^q = e$ imply $x = e$,
 for every $X \in \mathfrak{g} = \mathrm{Lie} G$ such that $X^{(p)} = X$, one has $X = 0$.
- ii bis) a) if $p = 0$: as in ii) a);
 b) if $p \neq 0$: every finite subgroup of $G(k)$ is a p -group,
 for every $X \in \mathfrak{g} = \mathrm{Lie} G$ such that $X^{(p)} = X$, one has $X = 0$.

Proof.

i) $\Leftrightarrow$ i bis), since the image of a group of multiplicative type is of multiplicative type (Exp. IX 2.7).

ii) $\Leftrightarrow$ ii bis). For if an ordinary finite group H has an order that is not a power of p , there are a prime number $q \neq p$ and an element x of H distinct from e such that $x^q = e$ (Sylow's theorem, cf. J.-P. Serre, *Corps locaux*, chap. IX § 2).

i) $\Rightarrow$ ii) follows from the following lemma:

Lemma 3.11. *Let G be an algebraic group defined over a field k .*

a) *If k contains the n th roots of unity, n an integer prime to p , one has:*

$$\mathrm{Hom}_{k\text{-gr}}(\mu_n, G) \simeq \mathrm{Hom}_{k\text{-gr}}(\mathbf{Z}/n\mathbf{Z}, G) \simeq {}_nG(k)$$

(points of order n of $G(k)$).

b) *If $p > 0$, $\mathrm{Hom}_{k\text{-gr}}(\mu_p, G) \simeq \{X \in \mathfrak{g} = \mathrm{Lie} G \mid X^{(p)} = X\}$.*

Indeed, to prove a), one notes that μ_n is then isomorphic over k to $(\mathbf{Z}/n\mathbf{Z})$, and b) follows from App. II 2.1.

ii) $\Rightarrow$ i bis). By Lemma 3.11, ii) is equivalent to the fact that for every prime number r , G contains no subgroups μ_r , which implies i bis) by:

Lemma 3.12. *Let G be a diagonalizable S -group, of finite type over S and distinct from the unit group. Then there are a prime number r and a subgroup of G isomorphic to $(\mu_r)_S$.*

Let $G = D_S(M)$, where M is a finitely generated abelian group, hence an extension of a free group M'' by a finite group M' . If $M'' \neq 0$, it is clear that M admits quotients isomorphic to $\mathbf{Z}/r\mathbf{Z}$ for every integer r . If $M'' = 0$, M' admits a quotient isomorphic to $\mathbf{Z}/r\mathbf{Z}$ for every prime number r dividing the order of M . One deduces the lemma by duality.

We have seen (Prop. 2.4) that a unipotent group satisfies the equivalent conditions of 3.10. The purpose of the following paragraph is to prove the converse.

4. A characterization of unipotent groups

As announced, we shall show that an algebraic group G defined over an algebraically closed field k , containing no nonzero subgroup of multiplicative type, is unipotent. In fact, it suffices that it contain no “elementary” subgroups of multiplicative type of certain very particular kinds, depending on the hypotheses made on G . Before stating the general theorem, let us study a few particular cases in detail.

4.1. Smooth connected affine algebraic groups. —

Proposition 4.1.1. *Let k be a field, G a smooth connected affine algebraic k -group, $\mathfrak{g} = \text{Lie } G$. Then the following properties are equivalent:*

- i) G is unipotent.
- ii) G has a central composition series whose successive quotients are forms of $\mathbf{G}_a$.
- iii) G has a central characteristic composition series whose successive quotients are forms of $(\mathbf{G}_a)^r$.
- iv) There is an integer $n > 1$ such that $G_{\bar{k}}$ contains no subgroup isomorphic to μ_n .
- v) Every maximal torus of G is the unit group.

Suppose moreover that G is an algebraic subgroup of a linear group $\text{GL}(n)$. Then the preceding conditions are also equivalent to:

- vi) $\mathfrak{g} \subset \mathfrak{gl}(n)$ consists of nilpotent endomorphisms.
- vii) $\mathfrak{g}$ is nilpotent, and its center contains no nonzero semisimple endomorphism.

Proof. ii) $\Rightarrow$ i) is clear, and i) $\Rightarrow$ iii) was seen in 3.9. The implication iii) $\Rightarrow$ ii) will follow from the following lemma:

Lemma 4.1.2. *Let k be a field, G an algebraic k -group that is a form of $(\mathbf{G}_a)^r$. Then:*

- a) G can be realized as an algebraic subgroup of the group $(\mathbf{G}_a)^n$ for a suitable integer n .
- b) G has a composition series whose successive quotients are forms of $\mathbf{G}_a$.

Indeed, by hypothesis, there is an extension k' of k such that $G_{k'}$ is isomorphic to $(\mathbf{G}_{a,k'})^r$. By the principle of finite extension (EGA IV 9.1.1), one can assume that k' is a finite extension of k . But then, for a), it suffices to consider the canonical closed immersion (EGA V⁽⁵⁾):

$$G \longrightarrow \prod_{k'/k} (G_{k'})/k' \xrightarrow{\sim} (\mathbf{G}_{a,k})^n \quad (\text{with } n = r \deg(k'/k)).$$

To prove b), in view of a), one can assume that G is a closed subgroup of $G' = (\mathbf{G}_a)^n$. If $G \neq 0$, there is a hyperplane $\mathfrak{h}$ of $\mathfrak{g}' = \text{Lie } G'$ that does not contain $\mathfrak{g}$. Let H be the vector subgroup $W(\mathfrak{h})$ of $W(\mathfrak{g}') = G'$. Since H is defined by one equation in G' , $H \cap G$ is defined by one equation in G , and one has the inequalities:

$$\begin{aligned} \dim G - 1 &\leq \dim(G \cap H) \\ &\leq \dim \text{Lie}(G \cap H) = \dim_k(\mathfrak{g} \cap \mathfrak{h}) \\ &= \dim_k(\mathfrak{g}) - 1 = \dim G - 1. \end{aligned}$$

Hence $\dim(G \cap H) = \dim \text{Lie}(G \cap H)$, and consequently $G \cap H$ is smooth. The group $G_1 = (G \cap H)^0$ is a smooth connected algebraic subgroup of G such that G/G_1 is smooth, connected, of dimension 1, hence is a form of $\mathbf{G}_a$ (4.1 i) $\Rightarrow$ iii)). One finishes by induction on the dimension of G .

Before continuing the proof of 4.1, note that the equivalence i) $\Leftrightarrow$ ii) and 2.3 bis imply the following corollary:

⁽⁵⁾**N.D.E.** Give another reference here. . .

Corollary 4.1.3. *If k is a perfect field, a smooth connected algebraic k -group is unipotent if and only if it has a composition series with successive quotients isomorphic to $\mathbf{G}_a$.*

Continuation of the proof of 4.1.

i) $\Rightarrow$ iv) by 2.4 i).

iv) $\Rightarrow$ v). By Exp. XIV 4.1, G has a maximal torus T defined over k . Now, if $r = \dim T$, $({}_nT)_{\bar{k}}$ is isomorphic to $(\mu_n)^r$. Hence $r = 0$.

v) $\Rightarrow$ i), as one observed in the proof of 3.8.

i) $\Rightarrow$ vi). By 3.4, G is in fact contained in an algebraic subgroup of $\mathrm{GL}(n)$ isomorphic to $\mathrm{Trigstr}(n)$, hence $\mathfrak{g}$ consists of nilpotent endomorphisms.

vi) $\Rightarrow$ vii). Indeed, $\mathfrak{g}$ is nilpotent by Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, chap. I § 4 cor. 3).

vii) $\Rightarrow$ v). Let T be a maximal subtorus of G (Exp. XIV 1.1), $\mathfrak{t}$ its Lie algebra. The embedding of G in $\mathrm{GL}(n)$ defines a representation of T that is necessarily semisimple (this is seen over an algebraic closure $\bar{k}$ of k , and one applies Exp. I 4.7.3). Thus, if $X \in \mathfrak{t}$, X is a semisimple endomorphism in $\mathfrak{gl}(n)$. One immediately sees that this implies that the map:

$$\mathrm{ad} X : Y \mapsto [X, Y]$$

is a semisimple endomorphism of $\mathfrak{gl}(n)$, hence of $\mathfrak{g}$. Since, moreover, this endomorphism is nilpotent, $\mathfrak{g}$ being nilpotent, $\mathrm{ad} X$ is zero, hence X is central. But then $\mathfrak{t}$ is central and consists of semisimple endomorphisms, hence is zero by hypothesis; a fortiori, T is the unit group.

Remark 4.1.4. a) We shall give later (4.3.1) an infinitesimal characterization of unipotent groups in characteristic $p > 0$, independent of an embedding in $\mathrm{GL}(n)$.

b) When k is perfect, conditions ii) and iii) of 4.1.1 simplify by virtue of the following lemma:

Lemma 4.1.5. *If k is a perfect field, every algebraic k -group G that is a form of $(\mathbf{G}_a)^r$ is isomorphic to $(\mathbf{G}_a)^r$.*

The lemma follows from 3.9 ter if the characteristic p of k is zero, and from 2.3 bis if $r = 1$. In the general case ($p > 0$), realize G as an algebraic subgroup of $(\mathbf{G}_a)^n$ for a suitable integer n (4.1.2), and argue by induction on the integer $n - r$. If $r = n$, one does indeed have $G = (\mathbf{G}_a)^r$. Otherwise, the quotient group $\mathbf{G}_a^n/G$ is a nonzero smooth connected unipotent group which, in view of 4.1.1 i) $\Rightarrow$ ii) and 2.3 bis, has a composition series with quotients isomorphic to $\mathbf{G}_a$. One deduces that there is a smooth connected algebraic subgroup G_1 of $(\mathbf{G}_a)^n$, containing G , such that $\mathbf{G}_a^n/G_1 = \mathbf{G}_a$. By induction, it suffices to show that G_1 is isomorphic to $(\mathbf{G}_a)^{n-1}$. Now it is immediate to check that a homomorphism from $\mathrm{Spec} k[X_1, \dots, X_n] = \mathbf{G}_a^n$ to $\mathbf{G}_a = \mathrm{Spec} k[T]$ is defined by an additive polynomial of the form:

$$\sum_{i,j} a_{i,j} X_i^{p^j}.$$

Since G_1 is smooth, the linear part of this polynomial is nonzero. After making a linear change in the coordinates X_i , we can assume that G_1 is an algebraic subgroup of $(\mathbf{G}_a)^n$ defined by the equation:

$$(*) \quad P(X) = -X_1 + \sum_{j=1}^q a_j X_1^{p^j} + Q(X_2, \dots, X_n) = 0,$$

$$\text{with } Q(X_2, \dots, X_n) = \sum_{\substack{i \geq 1 \\ j > 0}} b_{i,j} X_i^{p^j}.$$

Let us then proceed by induction on the degree of P . If $\deg P = 1$, it is clear that $G_1 \xrightarrow{\sim} \mathbf{G}_a^{n-1}$. Otherwise, since k is perfect, one has $Q(X) = Q_1(X)^p$, and we can define an endomorphism

v of $\mathbf{G}_a^n$ by the formulas:

$$X_i \mapsto X_i \quad \text{for } i > 1, \quad X_1 \mapsto \sum_{j=1}^q a_j^{1/p} X_1^{p^{j-1}} + Q_1(X).$$

It is clear that v induces an isomorphism on G_1 and that $v(G_1)$ has equation in $\mathbf{G}_a^n$:

$$(*)_1 \quad P_1(X) = -X_1 + \sum_{j=1}^q a_j^{1/p} X_1^{p^j} + Q_1(X_2, \dots, X_n) = 0.$$

Let us then distinguish two cases:

First case. $\deg(Q) = \deg P > p^q$. Then one has $\deg P_1 < \deg P$, and the induction hypothesis gives the result.

Second case. $\deg P = p^q$ ($a_q \neq 0$). One then has $\deg P_1 = \deg P = p^q$ and $\deg Q_1 < p^q$. (One cannot have $Q = 0$, for then G_1 would not be connected.) If the polynomial Q_1 has no linear part, one can repeat the preceding transformation. Continuing the process, one finally obtains an equation of the form:

$$(*)_s \quad P_s(X) = -X_1 + \sum_{j=1}^q a_j^{1/p^s} X_1^{p^j} + Q_s(X),$$

where $Q_s(X) = Q(X_2, \dots, X_n)^{1/p^s}$ is an additive polynomial with a nonzero linear part and, moreover, $\deg Q_s < p^q$. Suppose, for example, that the coefficient of X_2 in Q_s is nonzero, and let $-L$ be the linear part of $P_s(X)$. After making a linear change of coordinates, the equation of G_1 becomes:

$$P'(X) = -L + \sum_{j=1}^q a_j^{1/p^s} X_1^{p^j} + Q'(L, X_3, \dots, X_n),$$

where Q' is an additive polynomial, without linear part, and $\deg(Q') < p^q$. But then we are reduced to the first case⁽⁶⁾.

4.2. Radicial groups. —

Proposition 4.2.1. *Let G be a radicial algebraic group defined over a field k of characteristic $p > 0$. Then the following conditions are equivalent:*

- i) G is unipotent.
- ii) G has a central composition series with successive quotients isomorphic to α_p .
- iii) G has a central characteristic composition series with successive quotients isomorphic to $(\alpha_p)^r$.
- iv) $G_{\bar{k}}$ contains no subgroup isomorphic to μ_p .
- v) $\mathfrak{g} = \text{Lie } G$ is a unipotent Lie p -algebra (3.6).

Proof. iii) $\Rightarrow$ ii) $\Rightarrow$ i) is clear, i) $\Rightarrow$ iii) is 3.9 i), and i) $\Rightarrow$ iv) by 2.4 i).

We shall need the following lemma on abelian Lie p -algebras:

Lemma 4.2.2. *Let $\mathfrak{g}$ be an abelian Lie p -algebra, finite-dimensional over a perfect field k . Then $\mathfrak{g}$ can be written uniquely as the direct sum of a Lie p -subalgebra $\mathfrak{r}$ on which the p th power is bijective and a unipotent Lie p -subalgebra $\mathfrak{u}$ (3.6). (The algebra $\mathfrak{r}$ will be called the reductive part of $\mathfrak{g}$, and $\mathfrak{u}$ the unipotent part.) The formation of $\mathfrak{r}$ and $\mathfrak{u}$ is compatible with extension of the field k . If, moreover, k is algebraically closed, $\mathfrak{r}$ admits a basis e_i such that $e_i^{(p)} = e_i$.*

The proof of this lemma is easy and left to the reader (cf. Bourbaki, *Groupes et algèbres de Lie*, chap. I § 1 exercise 23). Let us simply say that $\mathfrak{u}$ is the kernel of a suitable iterate of the map $X \mapsto X^{(p)}$, and that $\mathfrak{r}$ is the image of the same iterate.

⁽⁶⁾**N.D.E.** By replacing X_1 by L .

This being so, let us prove iv) $\Rightarrow$ v). After extending the base field, one can assume that k is algebraically closed.

Let then X be an element of $\mathfrak{g}$, and $\mathfrak{h}$ the Lie p -subalgebra generated by X in $\mathfrak{g}$. The algebra $\mathfrak{h}$ is obviously commutative, and its reductive part (4.2.2) is zero, for otherwise, by *loc. cit.*, $\mathfrak{h}$ would contain an element $Y \neq 0$ such that $Y^{(p)} = Y$, and consequently (App. II 2.1 and 2.2) G would contain a subgroup isomorphic to μ_p , contrary to the hypothesis. Thus $\mathfrak{h}$, and consequently $\mathfrak{g}$, is a unipotent Lie p -algebra.

v) $\Rightarrow$ i). This is the least trivial implication of 4.2.1.

a) *Case where G is of height 1* (Exp. VIIA 4.1.3). Since G is radicial, it is affine, hence isomorphic to an algebraic subgroup of a linear group $\mathrm{GL}(V)$ (Exp. VIB § 11). This embedding identifies $\mathfrak{g}$ with a Lie p -subalgebra of $\mathrm{End}(V)$, the p th power of X in $\mathfrak{g}$ coinciding with the p th power of the endomorphism X (Exp. VIIA 6.4.4). Since $\mathfrak{g}$ is unipotent by hypothesis, $\mathfrak{g}$ is therefore an algebra of nilpotent endomorphisms of V , and by *Engel's theorem* (Bourbaki, *Groupes et algèbres de Lie*, chap. I § 4 th. 1) is a Lie subalgebra of the Lie algebra $\mathfrak{h}$ of the group of strict upper triangular matrices $\mathrm{Trigstr}(n)$ with respect to a suitable basis of V . Since G is of height 1, one then deduces from App. II 2.2 that G itself is an algebraic subgroup of $\mathrm{Trigstr}(n)$, hence is unipotent (3.5 v) $\Rightarrow$ i)).

b) *General case.* Let us proceed by induction on the height h of G^* . The case $h = 1$ has just been treated. Suppose that $h > 1$, and put $G' = {}_F G$ and $G'' = G/G'$. The group G' is of height 1, and $\mathrm{Lie} G' = \mathrm{Lie} G$ is unipotent, hence G' is unipotent by a). To show that G is unipotent, it therefore suffices to prove that G'' is unipotent (2.2). But G'' is of height $h - 1$, hence, by the induction hypothesis, it suffices to show that $\mathrm{Lie} G''$ is unipotent. Since iv) $\Rightarrow$ v), it suffices to show that $G''_{\bar{k}}$ contains no groups isomorphic to μ_p . Let therefore a subgroup of $G''_{\bar{k}}$ isomorphic to μ_p be given, and let H be its inverse image in $G'_{\bar{k}}$. Since the group G' is unipotent, we shall prove in § 5 that the extension:

$$e \longrightarrow G'_{\bar{k}} \longrightarrow H \longrightarrow \mu_p \longrightarrow e$$

is necessarily trivial (the proof given of this fact is independent of the results of the present paragraph). Thus the group μ_p lifts into H , but, being of height 1, it is necessarily contained in $G'_{\bar{k}} = {}_F G_{\bar{k}}$; hence a contradiction, since G' is unipotent.

4.3. Connected affine groups in characteristic $p > 0$. —

Proposition 4.3.1. *Let G be a connected affine algebraic group over a field k of characteristic $p > 0$. Then the following conditions are equivalent:*

- i) G is unipotent.
- ii) G has a composition series with successive quotients isomorphic to α_p and $\mathbf{G}_a$ (taken in this order).
- iii) G admits a characteristic composition series with successive quotients isomorphic to $(\alpha_p)^r$ and $(\mathbf{G}_a)^s$ (taken in this order).
- iv) $G_{\bar{k}}$ contains no subgroup isomorphic to μ_p .
- v) $\mathfrak{g} = \mathrm{Lie} G$ is unipotent (3.6).
- vi) $\mathfrak{g}$ is nilpotent, and the reductive part of the center of $\mathfrak{g}$ (4.2.2) is trivial.
- vi bis) $\mathfrak{g}$ is nilpotent, and every subgroup of multiplicative type of the neutral component of the center of G is zero.
- vi ter) G is nilpotent, and every subgroup of multiplicative type of the neutral component of the center of G is zero.

Proof. It is clear that iii) $\Rightarrow$ ii) $\Rightarrow$ i). To establish i) $\Rightarrow$ iii), we shall need the following lemma:

Lemma 4.3.2. *Let k be a field of characteristic $p > 0$, $n \in \mathbf{N}$, k' a radicial extension of k such that $(k')^{p^n}$ is contained in k ; for every k -prescheme X (resp. every k' -prescheme X'),*

*I.e. the least integer such that $F^h = \mathrm{id}_G$, cf. App. II 1.

denote by $X^{(p^n)}$ (resp. X'_φ) the k -prescheme obtained from X (resp. X') by the base change:

$$F^n : k \longrightarrow k, \quad x \longmapsto x^{p^n}, \quad (\text{resp. } \varphi : k' \longrightarrow k, \quad x' \longmapsto x'^{p^n}).$$

Then, for every k -prescheme X , there is a functorial isomorphism:

$$(X_{k'})_\varphi \xrightarrow{\sim} X^{(p^n)}.$$

Consequently, if X and Y are two k -preschemes such that there is a k' -isomorphism $u' : X_{k'} \xrightarrow{\sim} Y_{k'}$, then there is a k -isomorphism $v : X^{(p^n)} \xrightarrow{\sim} Y^{(p^n)}$. If, moreover, X and Y are endowed with structures of group k -preschemes, and if u' is a k' -homomorphism, then v is a k -homomorphism.

The lemma simply follows from the transitivity of base changes and the fact that the composite morphism $k \longrightarrow k' \xrightarrow{\varphi} k$ is equal to F^n .

Continuation of the proof of 4.3.1.

i) $\Rightarrow$ iii). Let us proceed by induction on $\dim G$. If $\dim G = 0$, since G is connected, it is radicial, and one applies 3.9 i). If $\dim G > 0$, there is an integer $m \geq 0$ such that the quotient $G/F^m G$ is a smooth group (App. II 3.1), obviously connected and nonzero. Applying 4.1.1 i) $\Rightarrow$ iii) to the latter, one sees that there is an algebraic subgroup G' of G that is characteristic and connected and such that the quotient $G'' = G/G'$ is a form of $\mathbf{G}_a^r$ ($r > 0$). By 4.1.5, if K is a perfect closure of k , one has $G''_K \xrightarrow{\sim} (\mathbf{G}_{a,K})^r$. Since G'' is of finite type over k , there is a finite radicial extension k' of k such that $G''_{k'} \xrightarrow{\sim} (\mathbf{G}_{a,k'})^r$ (Exp. VIB § 10). Let $n > 0$ be such that $(k')^{p^n} \subset k$. Keeping the notation of 4.3.2, one deduces that there is a k -isomorphism of algebraic groups:

$$(\mathbf{G}_{a,k})^r = (\mathbf{G}_{a,k'})^r_\varphi \xrightarrow{\sim} (G'')^{(p^n)}.$$

Consider then the Frobenius homomorphism relative to G'' (Exp. VIIA § 4)

$$F^n : G'' \longrightarrow (G'')^{(p^n)}.$$

Since G'' (hence also $(G'')^{(p^n)}$) is smooth over k , and F^n is radicial, F^n is an epimorphism for the fpqc topology, so that $(G'')^{(p^n)}$ is identified with $G''/F^n(G'')$. Finally, we have shown that $G''/F^n(G'')$ is isomorphic, as an algebraic group, to $(\mathbf{G}_a)^r$. The inverse image G'_n of $F^n(G'')$ in G is a connected characteristic subgroup of G of dimension strictly less than that of G , to which we can apply the induction hypothesis.

i) $\Rightarrow$ iv) by 2.4 i).

iv) $\Rightarrow$ i). Consider G as an extension of a smooth connected group G'' by a radicial group G' (App. II 3.1). The group G' is unipotent (4.2.1 iv) $\Leftrightarrow$ i)). It suffices to see that G'' is unipotent, and for this it suffices to show that $G''_{\bar{k}}$ contains no subgroup isomorphic to μ_p (4.1.1 i) $\Leftrightarrow$ iv)). Now, if $G''_{\bar{k}}$ contained a subgroup μ_p , the latter would lift into $G_{\bar{k}}$ by the result (5.1) already used and proved in § 5; hence a contradiction with iv).

i) $\Rightarrow$ v) by 3.7.

v) $\Rightarrow$ vi). Indeed, since $(\text{ad } X)^{p^r} = \text{ad}(X^{(p^r)})$ (VIIA 5.2), $\text{ad } X$ is nilpotent if $\mathfrak{g}$ is unipotent, hence $\mathfrak{g}$ is nilpotent by *Engel's theorem* (Bourbaki, *Groupes et algèbres de Lie*, chap. I § 4). Moreover, if $\mathfrak{g}$ is unipotent, the same is obviously true of its center, whose reductive part is then trivial (4.2.2).

vi) $\Rightarrow$ iv). Indeed, if $G_{\bar{k}}$ contains a subgroup isomorphic to μ_p , there is an element $X \neq 0$ of its Lie algebra such that $X^{(p)} = X$ (App. II 2.1), hence $X^{(p^r)} = X$ for every $r > 0$. Since $\text{ad } X$ is nilpotent because $\mathfrak{g}$ is nilpotent, and $(\text{ad } X)^{p^r} = \text{ad } X^{(p^r)}$, one necessarily has $\text{ad } X = 0$, hence X belongs to the reductive part of the center of $\mathfrak{g}_{\bar{k}}$; hence a contradiction with vi).

i) $\Rightarrow$ vi ter) follows from 2.4 i) and 3.5 i) $\Rightarrow$ iii).

vi ter) $\Rightarrow$ vi bis). Indeed, if G is nilpotent, so is its subgroup ${}_F G$. Moreover, it follows from App. II 2.2 that ${}_F G$ is nilpotent if and only if $\text{Lie } {}_F G = \text{Lie } G$ (Exp. VIIA) is nilpotent.

vi bis) $\Rightarrow$ vi). Let Z be the neutral component of the center of G , and let $\mathfrak{r}$ be the reductive component of the center of $\mathfrak{g}$. We must show that $\mathfrak{r} = 0$. Now it is immediate that $\mathfrak{r}$ is a characteristic Lie p -subalgebra of $\mathfrak{g} = \text{Lie } {}_F G$ (that is, stable under the functor $\underline{\text{Aut}}_{p\text{-Lie}}(\mathfrak{g})$); hence $\mathfrak{r}$ is the Lie algebra of a characteristic radicial subgroup R of ${}_F G$ (App. II 2.2). Moreover, it follows from the last assertion in 4.2.2 and from App. II 2.1 that R is a form of $(\mu_p)^r$. Since the group R is characteristic in ${}_F G$, which is itself a characteristic subgroup of G (App. II 1), R is a fortiori invariant in G , hence is central, since G is connected (Exp. IX 5.5). Thus, by hypothesis vi bis), R is zero, and therefore so is $\mathfrak{r}$.

4.4. Étale groups. — The following proposition is an easy consequence of Sylow's theorems and of the structure of finite q -groups (cf. J.-P. Serre, *Corps locaux*, chap. IX § 1).

Proposition 4.4.1. *Let G be a finite étale algebraic group defined over an algebraically closed field k . Then, for G to be unipotent, it is necessary and sufficient that for every prime number q distinct from the characteristic p of k , G contain no subgroup isomorphic to μ_q .*

4.5. Abelian varieties. — Let G be an abelian variety defined over an algebraically closed field k . Then the following conditions are equivalent:

- i) G is unipotent.
- ii) $G = 0$.
- iii) There is an integer n , prime to the characteristic p of k , such that G contains no subgroup isomorphic to μ_n .

Indeed, if G is an abelian variety of dimension d , one knows (cf. S. Lang, *Abelian varieties*, chap. IV § 3, th. 6) that the group ${}_n G(k)$ (n an integer prime to p) is isomorphic to $(\mathbf{Z}/n\mathbf{Z})^{2d}$, hence is isomorphic to $(\mu_n)^{2d}$. Hence iii) $\Rightarrow$ ii), and ii) $\Rightarrow$ i) $\Rightarrow$ iii) are obvious.

4.6. General case. — If G and H are two algebraic groups defined over an algebraically closed field k , we shall denote by $P(G, H)$ the property: “there is no algebraic subgroup of G isomorphic to H ”. One then obtains the following characterizations of unipotent groups:

Theorem 4.6.1. *Let G be an algebraic group defined over an algebraically closed field k of characteristic p . Then:*

- i) *If G is smooth, affine and connected:*

$$\begin{aligned} G \text{ is unipotent} &\iff \exists n > 1 \text{ such that } P(G, \mu_n) \text{ is true} \\ &\iff P(G, \mathbf{G}_m) \text{ is true.} \end{aligned}$$

- ii) *If G is smooth and connected:*

$$G \text{ is unipotent} \iff \exists n \text{ prime to } p \text{ such that } P(G, \mu_n) \text{ is true.}$$

- iii) *If G is smooth:*

$$G \text{ is unipotent} \iff \text{for every prime number } n \neq p, P(G, \mu_n) \text{ is true.}$$

- iv) *G is affine and connected, and $p > 0$:*

$$G \text{ unipotent} \iff P(G, \mu_p) \text{ is true.}$$

- v) *G is connected and $p > 0$:*

$$\begin{aligned} G \text{ unipotent} &\iff \exists n \text{ prime to } p \text{ such that } P(G, \mu_n) \text{ is true} \\ &\quad \text{and } P(G, \mu_p) \text{ is true.} \end{aligned}$$

- vi) *G any algebraic group:*

$$G \text{ is unipotent} \iff \text{for every prime number } n, P(G, \mu_n) \text{ is true.}$$

Proof. i) follows from 4.1.1, and iv) from 4.3.1. We shall prove vi); ii), iii), v) are proved similarly and are left to the reader.

Let therefore G be an algebraic group. If G is unipotent, $P(G, \mu_n)$ is true for every $n > 1$ (2.4 i)). Conversely, suppose that $P(G, \mu_n)$ is true for every prime number n , and let us show that G is unipotent. Let G^0 be the neutral component of G . If G^0 is smooth, it follows from a classical theorem of Chevalley* that G^0 is an extension of an abelian variety A by a smooth connected affine group L . If G is not smooth, which supposes $p > 0$ (Exp. VIB 1.6.1), there is an integer $n > 0$ such that $G'' = G^0 /_{F^n}(G)$ is smooth (App. II 3.1). Thus G'' is an extension of an abelian variety A by a smooth connected linear group L'' . Denote by L the inverse image of L'' in G^0 , which is again affine and connected, since $_{F^n}(G)$ is radicial. In all cases, G therefore has a composition series:

$$0 \subset L \subset G^0 \subset G$$

such that L is affine and connected, $G^0/L = A$ is an abelian variety, and G/G^0 is an étale group.

If $P(G, \mu_n)$ is true, a fortiori $P(L, \mu_n)$ is true, hence L is unipotent (4.1.1 and 4.3.1). If A is nonzero, there are a prime number n and a subgroup of A isomorphic to μ_n (4.5); by 5.1 below, this subgroup lifts into G , contradicting hypothesis $P(G, \mu_n)$; hence $A = 0$. Finally, if G/G^0 is not unipotent, there are an integer q and a subgroup of G/G^0 isomorphic to μ_q (4.4.1). One deduces, as above, that $P(G, \mu_q)$ is not true; hence G/G^0 is unipotent, and consequently the same is true of G .

5. Extension of a group of multiplicative type by a unipotent group

5.1. Statement of the theorem. —

Definition 5.1.0. Let k be a field, G an algebraic k -group. Following the terminology introduced by Rosenlicht (*Questions of rationality for solvable algebraic groups over non perfect fields*, Annali di Math. 61 (1963)), we shall say that G is “ k -solvable” if it satisfies the following equivalent conditions:

- i) G has a composition series with successive quotients isomorphic to $\mathbf{G}_a$.
- ii) G has a characteristic composition series (G_i) such that the successive quotients G_i/G_{i+1} are commutative and have a composition series with successive quotients isomorphic to $\mathbf{G}_a$.

The fact that i) $\Rightarrow$ ii) is proved in *loc. cit.*. In fact, for the composition series (G_i) , one can take the composition series introduced in the proof of 3.9 ii).

Theorem 5.1.1. Let k be a field, U a unipotent algebraic k -group, H a k -group of multiplicative type, E an algebraic k -group that is an extension of H by U , so that one has the exact sequence:

$$1 \longrightarrow U \longrightarrow E \longrightarrow H \longrightarrow 1.$$

Then:

- i) The extension E is trivial in each of the following cases:
 - a) k is algebraically closed.
 - b) k is perfect, and one of the groups U or H is connected.
 - c) U is k -solvable.
 - d) U is smooth and H is connected.
- ii) If H' and H'' are two liftings of H into E , then H' and H'' are conjugate by an element of $U(k)$ in each of the following cases:
 - a) k is algebraically closed and H is smooth.
 - b) k is algebraically closed and U is smooth.

*Sém. Bourbaki no. 145, 1956/57.

(We point out, during the proof, other cases where the conclusion of ii) is true without assuming that k is algebraically closed.)

If U is an algebraic group (resp. a commutative algebraic group) defined over a field k , we denote by $H^1(k, U)$ (resp. $H^i(k, U)$, $i \geq 0$) the pointed set (resp. the i th group) of Galois cohomology of k with values in U (cf. J.-P. Serre, *Cohomologie Galoisienne*, Lecture Notes Maths. no. 5, Springer).

If S is a prescheme, H a commutative group S -prescheme, G a group S -prescheme acting on H , we denote by $H^i(G, H)^{(7)}$ the i th Hochschild cohomology group of G with values in H (App. I 1).

To prove 5.1.1, we shall proceed in several steps.

5.2. Proof of 5.1.1 i) and ii) in the case where U is smooth and H is étale. —

Lemma 5.2.1. *With the notation of 5.1.1, if H is étale, the canonical morphism $E \rightarrow H$ has a section $s : H \rightarrow E$, defined over k , in the following cases:*

- a) k is algebraically closed.
- b) k is perfect, and U is smooth and connected.
- c) U is “ k -solvable”.

a) simply follows from the fact that $E(k) \rightarrow H(k)$ is surjective. One has b) $\Rightarrow$ c) by 4.1.2 b) and 2.3 (bis). It therefore suffices to treat case c). Now let x be a point of H ; x is therefore a subset of H that is both open and closed, and the induced subscheme is isomorphic to $\text{Spec } K$, where K is a finite separable extension of k . Let X be the inverse image in E of the scheme x . The K -scheme X is a principal homogeneous K -space under the group U_K . But U_K has a composition series with successive quotients isomorphic to $(\mathbf{G}_a)_K$, hence $H^1(K, U_K) = 0$ (J.-P. Serre, *Cohomologie Galoisienne*, chap. III, prop. 6), and consequently X is trivial, hence has a rational point over K . One thus obtains a k -section of $E \rightarrow H$ over x , for every point x of H , whence the existence of a section $H \rightarrow E$.

Lemma 5.2.3. ⁽⁸⁾ *With the notation of 5.1.1, suppose that H is étale. Then:*

- i) *The extension E is trivial in each of the following cases:*
 - a) U is commutative and $E \rightarrow H$ has a section.
 - b) k is algebraically closed.
 - c) k is perfect and U is connected.
 - d) U is “ k -solvable”.

Moreover, in each of the above cases, two liftings H' and H'' of H into E are conjugate by an element of $U(k)$.

- ii) *Let R be the k -functor $((\text{Sch})/k)^\circ \rightarrow (\text{Ens})$ such that, for every k -prescheme S , $R(S)$ is the set of liftings of H_S into E_S . Then, if U is commutative, R is representable by a nonempty affine k -scheme. The group U acts by inner automorphisms on R , and this action makes R a homogeneous space under U (for the fpqc topology (cf. Exp. IV)).*

Proof of i). We shall reduce cases b), c) and d) to case a).

Case b). Since U has a characteristic composition series with commutative successive quotients (3.5 i) $\Leftrightarrow$ iv)), one immediately reduces to the case where U is commutative. Moreover, $E \rightarrow H$ has a section by 5.2.1 a), and we are reduced to case a). *Case d).* One proceeds in the same way, using 5.1.0 ii) and 5.2.1 c).

Case c). Let E_{red} be the associated reduced subgroup of E . Since H is smooth, E_{red} is an extension of H by $U' = U \cap E_{\text{red}}$, and one can replace E by E_{red} . But U , hence also U' , is connected, and it is clear that U' is the neutral component of E_{red} , hence is smooth. But

⁽⁷⁾**N.D.E.** One has replaced H_0^i by H^i , to agree with the notation of App. I and Exp. I.

⁽⁸⁾**N.D.E.** There is no no. 5.2.2.

then U' is smooth and connected, k is perfect, hence U' is k -solvable (4.1.4 b)), and we are reduced to case d).

The preceding reductions show that in cases b), c) and d), one can assume that U has a characteristic composition series (U_i) such that U_i/U_{i+1} is commutative and such that the maps $U_i(k) \rightarrow (U_i/U_{i+1})(k)$ are surjective (in cases c) and d), this last point follows from $H^1(k, \mathbf{G}_a) = 0$). An immediate dévissage then shows that it suffices to prove the conjugacy of two liftings H' and H'' into E when U is commutative. Thus it suffices to prove i) a). In this case, the triviality of the extension E is assured if $H^2(H, U) = 0$ (App. I 3.1), and the conjugacy of H' and H'' if $H^1(H, U) = 0$. Now we have the following lemma:

Lemma 5.2.4. *Let S be a prescheme, U a commutative group S -prescheme, H an étale finite group S -prescheme of rank n , acting on U . Then the groups $H^i(H, U)$ ($i > 0$) are annihilated by n in the following two cases:*

- a) H is a constant S -group (Exp. I 4.1).
- b) S is the spectrum of a field.

Proof of a). By hypothesis, the group H is the constant group associated with an ordinary group $\{H\}$ of order n . It is then clear⁽⁹⁾ that $H^i(H, U)$ is isomorphic to the i th cohomology group $H^i(\{H\}, U(S))$ of the group $\{H\}$, with values in the ordinary group $U(S)$, and it is classical that these groups are annihilated by n (J.-P. Serre, *Corps locaux*, Chap. VIII, prop. 4 cor. 1).

Proof of b). Let $x \in H^i(H, U)$ ($i > 0$) be represented by an i -cocycle $f : H^{(i)} \rightarrow U$ (where $H^{(i)}$ denotes the product, over k , of i copies of H). If K is a finite Galois extension of k splitting H , it follows from a) that nf_K is a coboundary. More precisely, an easy calculation shows that if one defines the K -morphism $F_K : H_K^{(i-1)} \rightarrow U_K$ by the formula:

$$F_K(h_1, \dots, h_{i-1}) = \sum_{h \in H(K)} f_K(h_1, \dots, h_{i-1}, h),$$

one has, up to sign:

$$d(F_K) = nf_K \quad (d \text{ the coboundary operator}).$$

Now an immediate Galois argument shows that F_K comes from a k -morphism $F : H^{(i-1)} \rightarrow U$, and consequently one has $d(F) = nf$.

Corollary 5.2.4 bis. *With the notation of 5.2.4, suppose moreover that U is flat and of finite presentation over S , with unipotent fibers, and that n is prime to the residue characteristics of S . Then, in cases a) and b) above, one has $H^i(H, U) = 0$ for $i > 0$.*

It suffices to show that raising to the n th power in U is an isomorphism, for this will imply that multiplication by n in $H^i(H, U)$ is both an isomorphism and the zero morphism, hence $H^i(H, U) = 0$. Now, under the hypotheses made on U , it suffices to check that raising to the n th power is an isomorphism on the fibers of U (EGA IV 17.9.5), reducing us to the case where S is the spectrum of a field k of characteristic p . Since $(n, p) = 1$, raising to the n th power in U is an étale morphism (Exp. VII), and it is a monomorphism (2.4 i)), hence an open immersion (EGA IV 17.9.1). This already proves that the restriction to the neutral component U^0 is an isomorphism; since U/U^0 is a finite p -group, one has finished.

This completes the proof of 5.2.3 i) a), since H , being an étale group of multiplicative type, is of order prime to p .

Proof of 5.2.3 ii). It is clear that R is a sheaf for the fpqc topology. Moreover, in view of descent for affine schemes, the assertions of 5.2.3 ii) are local for the fpqc topology. We can therefore assume that k is algebraically closed. The group H is then split and the extension E is trivial (i b)); let H' be a lifting of H into E . For every k -prescheme S and every lifting

⁽⁹⁾**N.D.E.** This is Proposition III.6.4.2 of the book by M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson & North-Holland (1970).

H'' of H_S into E_S , H'_S and H'' are conjugate by an element of $U(S)$, since $H^1(H_S, U_S) = 0$ (5.2.4 bis). Let then $U^{H'}$ be the sheaf of invariants of U under H' , which is representable by an algebraic subgroup of U (Exp. VIII 6.5 d)). It follows from the preceding remarks that the k -morphism:

$$U \longrightarrow R, \quad u \longmapsto \text{int}(u)H' \quad (u \in U(S))$$

defines a k -isomorphism $U/U^{H'} \xrightarrow{\sim} R$. This proves the representability of R and the fact that R is affine (2.1).

Remark 5.2.5. One can show that the assertions of 5.2.3 ii) are still true when U is not commutative, but we shall not need this to prove 5.1.1.

5.3. Study of the case where H is smooth. —

Proposition 5.3.1. *The assertions in 5.2.3 i) remain true when one replaces the hypothesis “ H étale” by “ H smooth”.*

Proceeding as in the proof of 5.2.3 i), one reduces to the case where U is moreover commutative.

Let $\mathbf{N}'$ be the set of integers > 0 prime to p , ordered by divisibility. For every $n \in \mathbf{N}'$, ${}_nH^0$ is an étale group, and the family of the ${}_nH^0$ ($n \in \mathbf{N}'$) is schematically dense in H , since H is smooth (Exp. IX 4.10). Denote by E_n the inverse image of ${}_nH$ in E , so that E_n is an extension of ${}_nH$ by U ; finally, let R_n be the k -functor of liftings of ${}_nH$ into E_n (cf. 5.2.3 ii)). If n divides m , it is clear that one has a natural k -morphism $R_m \longrightarrow R_n$, so that the R_n form a projective system of k -functors. Since R_n is representable by a nonempty affine k -scheme (5.2.3 ii)), and since a filtered inductive limit of nonzero rings is nonzero, the functor $R = \varprojlim R_n$ is representable by a nonempty affine k -scheme (EGA IV 8 and 1.9.1). There is therefore an extension K of k and a point $u \in R(K)$. The image u_n of u in $R_n(K)$ corresponds to a lifting H'_n of $({}_nH)_K$ into $(E_n)_K$. By construction, $H'_n = {}_n(H'_m)$ if n divides m . Put $U_n = (U_K)^{H'_n}$. The choice of H'_n allows one to identify $(R_n)_K$ with U_K/U_n . But the family of the H'_n is increasing and filtered, hence the family of the U_n is decreasing and filtered and consequently is stationary for sufficiently large n (U_K is noetherian). It follows that the family of the $(R_n)_K$ is stationary, and consequently the same is true of the family of the R_n . In brief, one has $R_n = R$ for sufficiently large n .

Under the hypotheses made, it follows from 5.2.3 i) that $R_n(k)$ is nonempty. One can therefore find a coherent system of liftings H'_n of ${}_nH$ for $n \in \mathbf{N}'$. Denote by H' the smallest closed subscheme of E containing H'_n for every n (Exp. VIB §7). The argument made in Exp. XV 4.6 shows that H' is a smooth commutative algebraic group whose formation commutes with extension of the base field. To show that H' is a lifting of H into E , we can therefore assume that k is algebraically closed. By BIBLE 4 th. 4, H' is then the direct product of a (smooth) group of multiplicative type M and a unipotent group V . The groups H'_n are then necessarily contained in M (2.4), and in view of the definition of H' this implies $H' = M$. Thus H' is of multiplicative type and consequently $H' \cap U = 0$. The morphism $H' \longrightarrow H$ is therefore a monomorphism; moreover, it follows from the density theorem (Exp. IX 4.10) that it is an epimorphism, so it is indeed an isomorphism.

Let now H' and H'' be two liftings of H into E . For every $n \in \mathbf{N}'$, denote by $T_n = \text{Transp}({}_nH', {}_nH'')$ the transporter of ${}_nH'$ into ${}_nH''$, which is representable by a closed subscheme of U (Exp. VIII 6.5 e)). The T_n form a decreasing filtered family of closed subschemes of U , nonempty by 5.2.3 i a). Let T be the stationary value. Under the hypotheses of 5.2.3 i), $T_n(k)$ is nonempty. There is therefore an element u of $U(k)$ such that ${}_nH'' = \text{int}(u){}_nH'$ for every $n \in \mathbf{N}'$. But then $H'' = \text{int}(u)H'$ (Exp. IX 4.8 b).

5.4. Study of the case where U is radicial. —

Proposition 5.4.1. *If k is a perfect field of characteristic $p > 0$, and if U is radicial, the extension E of 5.1.1 is trivial.*

Using a characteristic composition series of U , we can restrict ourselves to the case where U is equal to $(\alpha_p)^r$ (3.9).

It follows from App. II 2.2 and 2.1 that one has isomorphisms of k -functors:

$$\underline{\mathrm{Aut}}_{k\text{-gr}}(\alpha_p)^r \xrightarrow{\sim} \underline{\mathrm{Aut}}_{p\text{-Lie}}(\mathrm{Lie}(\alpha_p)^r) \xrightarrow{\sim} \mathrm{GL}(\mathrm{Lie}(\alpha_p)^r).$$

Consider then a k -vector space V of rank r , the vector group k -scheme $W(V)$ (Exp. I 4.6), and identify $(\alpha_p)^r$ with ${}_FV$. The preceding remarks then show that the action of H on ${}_FV$, defined by the extension E , comes from a linear representation v of H in V . Consider then the exact sequence:

$$(*) \quad 0 \longrightarrow {}_FV \longrightarrow V \xrightarrow{F} V^{(p)} \longrightarrow 0.$$

where F is the Frobenius morphism. Then $(*)$ is an exact sequence of H -groups, provided that H acts on the factor $V^{(p)}$ through the linear representation $H \xrightarrow{v} \mathrm{GL}(V) \xrightarrow{F} \mathrm{GL}(V^{(p)})$.

Since the field k is perfect, the morphism $F : V \longrightarrow V^{(p)}$ induces a surjective map $V(k) \longrightarrow V^{(p)}(k)$. It then follows from App. I 2.1 that the exact sequence $(*)$ defines an exact sequence:

$$(**) \quad H^1(H, V^{(p)}) \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(H, {}_FV) \longrightarrow \mathrm{Ext}_{\mathrm{alg}}(H, V).$$

Let us show that $\mathrm{Ext}_{\mathrm{alg}}(H, V) = 0$. Let therefore E' be an algebraic group that is an extension of H by V :

$$1 \longrightarrow V \longrightarrow E' \xrightarrow{h} H \longrightarrow 1.$$

The scheme E' is a torsor with base H and group $\mathbf{G}_a^r$, hence defines an element of $H^1(H, \mathcal{O}_H^r)$ (in the sense of coherent sheaf cohomology). Since H is affine, one has $H^1(H, \mathcal{O}_H^r) = 0$ (EGA III §1). This says that $E' \longrightarrow H$ has a section. Consequently, the group $\mathrm{Ext}_{\mathrm{alg}}(H, V)$ is isomorphic to $H^2(H, V)$ (App. I 3.1). Now $H^i(H, V) = H^i(H, W(V)) = 0$ for $i > 0$ (Exp. IX 3.1). One then concludes, by the exact sequence $(**)$, that $\mathrm{Ext}_{\mathrm{alg}}(H, {}_FV) = 0$, hence that E is a trivial extension.

5.5. Proof of 5.1.1 i). —

If k has characteristic 0, U is k -solvable (4.1.3) and H is smooth; the fact that E is a trivial extension therefore follows from 5.3.1 and 5.2.3 d). One proves in the same way that two liftings of H into E are conjugate by an element of $U(k)$.

Henceforth, we therefore assume that k is a field of characteristic $p > 0$.

Proof of i) b): Case where k is perfect, U connected.

We shall reduce to the case where U is radicial. For this, note that since k is perfect, H_{red} is smooth; let E' be its inverse image in E . It then follows from 5.3.1 and 5.2.3 i) c) that the extension:

$$1 \longrightarrow U \longrightarrow E' \longrightarrow H_{\mathrm{red}} \longrightarrow 1$$

is trivial. Let H_1 be a lifting of H_{red} into E . By App. II 3.1, there is an integer $n > 0$ such that $E^{(n)} = E/{}_{F^n}(E)$ is smooth; let E'' be the algebraic subgroup of E generated by H_1 and ${}_{F^n}(E)$ (i.e. the inverse image in E of the image of H_1 in $E^{(n)}$). Denote by H'' the image of E'' in H . Then I say that $H'' = H$. Indeed, denote by R the image of ${}_{F^n}(E)$ in H , so that H'' is generated by R and H_{red} . The group H/R is a quotient of $E^{(n)}$, hence is smooth; consequently the canonical morphism

$$H_{\mathrm{red}} \longrightarrow H/R$$

is an epimorphism, hence H is generated by R and H_{red} , hence is equal to H'' . One thus obtains an exact sequence:

$$(\dagger) \quad 1 \longrightarrow U'' = U \cap E'' \longrightarrow E'' \longrightarrow H \longrightarrow 1.$$

But E'' has the same underlying space as H_1 , hence U'' is radicial. Moreover, it is clear that it suffices to prove that the extension $(\dagger)$ is trivial, which follows from 5.4.1.

Proof of i) b): Case where k is perfect, H connected.

The group U is an extension of an étale group by its neutral component U^0 . The case where U is connected having just been treated, it suffices to examine the case where U is étale. One then has the more precise lemma:

Lemma 5.5.1. *With the notation of 5.1.1, suppose moreover that U is étale. Then:*

- i) *If H is connected, there is a unique lifting of H into E , namely the neutral component E^0 of E .*
- ii) *If k is algebraically closed, E is trivial, and two liftings of H into E are conjugate by an element of $U(k)$.*

i) Since formation of the neutral component commutes with extension of the base field, we can restrict ourselves to the case where k is algebraically closed. If H is a torus, E is trivial (5.3.1 and 5.2.3 i b)), and it is clear that E^0 is the unique lifting of H . This already proves that in the general case, $E^0 \cap U$ is radical; since it is moreover étale (U being étale), it is zero. The morphism $E^0 \rightarrow H$ is therefore a monomorphism, flat (since E^0 is open in E) and surjective (H is connected), hence is an isomorphism. If now H' is another lifting of H , H' is connected, hence contained in E^0 and consequently is equal to E^0 .

ii) Let H^0 be the neutral component of H . By i), E^0 is the unique lifting of H^0 into E . Put $E' = E/E^0$, $H' = H/H^0$, so that one has the extension:

$$1 \rightarrow U \rightarrow E' \rightarrow H' \rightarrow 1.$$

Since H' is étale, this extension is trivial (5.2.3 i b)). If H'_1 is a lifting of H' into E' , and H_1 its inverse image in E , it is clear that H_1 lifts H into E . If H_2 is a second lifting of H into E , it contains E^0 ; by 5.2.3 i b), the image of H_2 in E' is conjugate to H'_1 by an element u of $U(k)$, whence immediately $H_2 = \text{int}(u)H_1$.

Remark 5.5.2. Under the hypotheses of 5.5.1 i), it is easy to see that E^0 centralizes U .

Proof of 5.1.1 i) a). Using the composition series

$$1 \rightarrow U^0 \rightarrow U \rightarrow U/U^0 \rightarrow 1,$$

i) a) follows from the conjunction of i) b) and 5.5.1 ii).

Before proving 5.1.1 c) and d), we shall first establish 5.1.1 ii).

Proof of 5.5.1 ii) a). For lack of a satisfactory general statement, we shall describe a certain number of cases where, when H is smooth, two liftings of H into E are conjugate:

Proposition 5.6.1. *With the notation of 5.1.1, suppose moreover that H is smooth, and let H' and H'' be two liftings of H into E . Then H' and H'' are conjugate by an element of $U(k)$ in each of the following cases:*

- a) *k is algebraically closed.*
- b) *U is commutative.*
- c) *k is perfect and U is connected.*
- d) *U is k -solvable.*
- e) *The centralizer of H' in U is k -solvable.*
- f) *The group of multiplicative type H is split by a finite Galois extension K of k of degree prime to p .*

Proof. a), b), c), d) follow from 5.3.1.

e) Let Z be the centralizer of H' in U , and T the transporter of H' into H'' . By a), T is nonempty, hence T is a principal homogeneous space under Z , and the hypothesis made on Z implies that it is trivial.

f) Proceeding as in 5.3.1, one sees that it suffices to consider the case where H is étale. Suppose first that H is diagonalizable, defined by the ordinary group M , of order m prime to p . The datum of the two liftings H' and H'' defines a 1-cocycle of M with values in $U(k)$, that is, a map h from M into $U(k)$ such that $h(mn) = h(m)^m h(n)$ for every pair m, n of

elements of M . The groups H' and H'' are conjugate by an element of $U(k)$ if and only if there exists $a \in U(k)$ such that

$$h(m) = a^{-1}({}^ma).$$

Now the abstract group $U(k)$ has a composition series with successive quotients commutative and annihilated by a power of p (one may assume $p > 0$ in view of 5.6.1 c)). One immediately deduces in this case that h is a coboundary.

Let us now examine the general case. Denote again by Z the centralizer of H' in U , and by T the transporter of H' into H'' , which is a torsor under Z . By hypothesis, there is a finite Galois extension K of k , with Galois group G , of order prime to p , which splits H . By the study made above, H'_K and H''_K are conjugate by an element of $U(K)$, hence T_K is trivial. Consequently T is defined by an element of $H^1(G, Z)$. For the same reasons as above, the hypothesis made on G implies that $H^1(G, Z) = e$, hence T is trivial.

5.7. Proof of 5.1.1 ii) b). —

Lemma 5.7.1. *Let S be a prescheme, G a group S -prescheme, separated and smooth over S , and H an S -group of multiplicative type acting on G . Then the S -functor $Z = G^H$ of invariants of G under H is representable by a subgroup S -prescheme of G , closed and smooth over S .*

The fact that G^H is representable by a closed subgroup prescheme of G follows from Exp. VIII 6.5 d). Consider then the semidirect product $K = G \times_S H$. The centralizer of H in K is then equal to $Z \times_S H$. To prove that Z is smooth, we must check that if S is affine, if S_0 is a closed subscheme of S defined by an ideal of square zero, and if u_0 is an element of $Z(S_0)$, then there is an element u of $Z(S)$ lifting u_0 . Now since G is smooth over S , there is an element u of $G(S)$ lifting u_0 . Let i be the canonical immersion of H into K , and consider the two group S -morphisms

$$H \rightrightarrows K, \quad i \text{ and } \text{int}(u)i = j.$$

Since u_0 belongs to $Z(S_0)$, one has $i_{S_0} = j_{S_0}$. By Exp. IX 3.2, there exists $v \in K(S)$ lifting the identity section of $K(S_0)$ and such that

$$i = \text{int}(v)j = \text{int}(vu)i.$$

Thus $vu = (u', v')$ belongs to $(Z \times_S H)(S)$. Thus u' belongs to $Z(S)$ and lifts u_0 .

Lemma 5.7.2. *Let k be a field, H an algebraic k -group, and let $P(H)$ be the following property:*

“for every smooth unipotent k -group U , and for every extension E of H by U , two liftings of H into E are conjugate by an element of $U(k)$ ”.

Then if H is an algebraic group that is an extension of a group of multiplicative type H'' by a group of multiplicative type H' , and if $P(H')$ and $P(H'')$ are true, $P(H)$ is true.

Indeed, let U be a smooth unipotent k -group, E an extension of H by U , H_1 and H_2 two liftings of H into E , and H'_1 and H'_2 the corresponding liftings of H' . Since $P(H')$ is true, there exists $u \in U(k)$ such that $H'_2 = \text{int}(u)H'_1$. Replacing H_1 by $\text{int}(u)H_1$, we can assume $H'_1 = H'_2$, which we allow ourselves to denote simply by H' . Let $E' = \underline{\text{Centr}}_E H'$, which is equal to

$$U^{H'} \cdot H_1 = U^{H'} \cdot H_2.$$

By 5.6.1, $U^{H'} = U'$ is smooth. Consider then the extension

$$1 \longrightarrow U' \longrightarrow E' \longrightarrow H \longrightarrow 1.$$

By construction H' is central in E' , hence invariant. Passing to the quotient, one obtains the exact sequence:

$$1 \longrightarrow U' \longrightarrow E'' \longrightarrow H'' \longrightarrow 1.$$

Since U' is smooth, and since $P(H'')$ is true, the two images of H_1 and H_2 in E'' are conjugate by an element u of $U'(k)$, but then $H_1 = \text{int}(u)H_2$.

To prove 5.1.1 ii) b), note then that, since k is algebraically closed, H has a composition series whose successive quotients are smooth or isomorphic to μ_p when $p > 0$. By repeated use of 5.7.2, we are reduced to the case where H is smooth or equal to μ_p . In the first case, it suffices to apply 5.1.1 ii) a). There remains the case $H = \mu_p$. Since U is smooth, U has a characteristic composition series with successive quotients étale or isomorphic to $(\mathbf{G}_a)^r$ (3.9). If U is étale, one applies 5.5.1. There remains finally the case $H = \mu_p$, $U = \mathbf{G}_a^r$.

We must show that $H^1(\mu_p, U) = 0$. The method used in 5.4.1 no longer applies here, for μ_p does not in general act linearly on $(\mathbf{G}_a)^r$. Let us fix the notation: H' denotes a lifting of H into E , $\mathfrak{e} = \text{Lie } E$, $\mathfrak{u} = \text{Lie } U$, $\mathfrak{h} = \text{Lie } H$, $\mathfrak{h}' = \text{Lie } H'$. Let X be a nonzero element of $\mathfrak{h}$ such that $X^{(p)} = X$ (App. II 3.1), and let X' be its lifting in $\mathfrak{h}'$.

Since μ_p is a radicial group of height 1, there is a one-to-one correspondence between the set of liftings of H into E and the set A of $Y \in \mathfrak{e}$ such that $Y^{(p)} = Y$ and which project onto X (App. II 2.2). Likewise, if $Y \in A$ corresponds to the lifting H'' of H into E , and if $u \in U(k)$, then $\text{int}(u)H' = H''$ if and only if $\text{Ad}(u)X = Y$. Let therefore B be the subset of $Y \in \mathfrak{e}$ of the form $\text{Ad}(u)X$, where $u \in U(k)$. One obviously has $B \subset A$, and everything reduces to showing that $A = B$ if k is algebraically closed.

a) Study of A . Since $\mathfrak{u}$ is commutative and normalized by $\mathfrak{h}$, the Jacobson formula (Exp. VIIA 5.2) gives simply here, for $u \in \mathfrak{u}$:

$$(X' + u)^{(p)} = X'^{(p)} + u^{(p)} + (\text{ad } X')^{p-1}(u) = X' + (\text{ad } X')^{p-1}(u),$$

so that $X' + u \in A \Leftrightarrow u = (\text{ad } X')^{p-1}(u)$.

Now let $\mathfrak{u} = \coprod_{n \in \mathbf{Z}/p\mathbf{Z}} \mathfrak{u}_n$ be the canonical decomposition of $\mathfrak{u}$ under the action of H' , which one can also write:

$$\mathfrak{u} = \mathfrak{u}_0 \oplus \coprod_{n \in (\mathbf{Z}/p\mathbf{Z})^\times} \mathfrak{u}_n.$$

If $u \in \mathfrak{u}_0$, one has $\text{ad } X'(u) = 0$. If $u \in \mathfrak{u}_n$ ($n \in (\mathbf{Z}/p\mathbf{Z})^\times$), one has $(\text{ad } X')^{p-1}(u) = u$. Finally, $Y = X' + u$ is an element of A if and only if $u \in \coprod_{n \in (\mathbf{Z}/p\mathbf{Z})^\times} \mathfrak{u}_n$. Let us retain that A is the set of rational points over k of an irreducible subscheme of $W(\mathfrak{e})$ of dimension equal to $\text{rk } \mathfrak{u} - \text{rk } \mathfrak{u}_0 = \text{rk } \mathfrak{u} - \text{rk } \text{Centr}_{\mathfrak{u}}(X')$.

b) Study of B . We shall need the following lemma:

Lemma 5.7.3. (Rosenlicht). *Let U be a unipotent algebraic group over a field k acting on a quasi-affine k -scheme X . Then the orbit of every point $x \in X(k)$ is closed in X (by orbit of x we mean the subset of $\text{ens}(X)$ that is the image of G under the morphism $g \mapsto g \cdot x$).*

By functoriality, G acts on the affine envelope of X (that is, $\text{Spec } \Gamma(X, \mathcal{O}_X)$), which allows us to assume that X is affine. One can then assume that k is algebraically closed, X reduced and U smooth (note that U_{red} acts on X_{red} if k is perfect). Let Y be the schematic image of G (EGA I 9.5.1) under the morphism $g \mapsto g \cdot x$, which is a closed reduced subscheme of X on which G acts. It follows easily from EGA IV 1.8.6 that the orbit of x is an open subset Z of Y , dense in Y . We must show that $Z = \text{ens}(Y)$. Let F be the reduced closed subscheme of Y having $Y \setminus Z$ as underlying space. One therefore has $F = V(J)$, where J is a nonzero ideal of $\Gamma(Y, \mathcal{O}_Y)$. Since G is smooth, G acts on F , hence on J and consequently (3.2) $J^G \neq 0$. If a is a nonzero element of J^G , a is necessarily constant on the orbit Z , hence is constant on Y , since Z is dense in Y . But then the ideal J contains k and $F = \emptyset$.

This being so, let us apply the preceding lemma to the group U acting on the affine space $W(\mathfrak{e})$ through the adjoint representation. One obtains that the orbit of X' is the underlying set of a closed subscheme of $W(\mathfrak{e})$. Moreover, the stabilizer Z of X' is the centralizer of X' in U , and one has a closed immersion:

$$U/Z \longrightarrow W(\mathfrak{e}).$$

Let us retain that the orbit of X' is the underlying space of a closed subscheme of $W(\mathfrak{e})$ of dimension equal to $\dim U - \dim Z$.

c) End of the proof of 5.1.1 ii) b). When k is algebraically closed, the canonical map $U(k) \rightarrow (U/Z)(k)$ is surjective, so that, by point b) above, B is the set of rational points of a closed subscheme of $W(\mathfrak{e})$ of dimension $\dim U - \dim Z$. In view of point a), to prove that $A = B$, it then suffices to show that one has:

$$\mathrm{rk} \, u - \mathrm{rk} \, \mathrm{Centr}_u(X') \leq \dim U - \dim \mathrm{Centr}_U(X').$$

Now since U is smooth, one has $\dim U = \mathrm{rk} \, u$. On the other hand, one has (Exp. II 5.3.3):

$$\dim \mathrm{Centr}_U(X') \leq \mathrm{rk} \, \mathrm{Centr}_u(X'),$$

whence the result (note that one has in fact $\dim \mathrm{Centr}_U(X') = \mathrm{rk} \, \mathrm{Centr}_u(X')$, which proves anew that Z is smooth (5.7.1)).

5.8. End of the proof of 5.1.1 i). —

It remains for us to prove i) c) and i) d).

Proof of 5.1.1 i) d) (U smooth, H connected). We shall need the following lemma:

Lemma 5.8.1. *With the notation of 5.1.1, suppose that U is smooth and H radicial, and let H_1 be an algebraic subgroup of H having a lifting H'_1 into E . Then H has a lifting H' into E containing H'_1 .*

By induction on the height of H/H_1 , one can assume that H/H_1 has height 1. Let $C' = \mathrm{Centr}_E(H'_1)$. I say that the canonical morphism $C' \rightarrow H$ is an epimorphism. To establish this point, we can assume that k is algebraically closed; but then the extension E is trivial (5.1.1 i a)); let H'' be a lifting of H into E , and H''_1 the inverse image of H_1 in H'' . The groups H'_1 and H''_1 are two liftings of H_1 into E , hence are conjugate by an element of $U(k)$, since k is algebraically closed and U smooth (5.1.1 ii b)). It is then clear that to prove the assertion about C' , it suffices to prove it for $C'' = \mathrm{Centr}_E(H''_1)$. But in this case, C'' contains H'' , and the property is clear. Moreover, it follows from 5.7.1 that $C \cap U$ is smooth. It is then clear that we can replace E by C , hence assume that H'_1 is central. But then, passing to the quotient by H'_1 , we can assume $H_1 = 0$ and H of height 1.

Since U is smooth, one has the exact sequence of Lie p -algebras (App. II 3.2):

$$(*) \quad 0 \rightarrow \mathrm{Lie} \, U \rightarrow \mathrm{Lie} \, E \rightarrow \mathrm{Lie} \, H \rightarrow 0.$$

In view of App. II 2.2, saying that E is trivial is equivalent to saying that $(*)$ is a trivial extension of Lie p -algebras. Suppose $H \neq 0$ (hence $\mathrm{Lie} \, H \neq 0$), and suppose that we have found a Lie p -subalgebra $\mathfrak{h}_1$ of $\mathfrak{h} = \mathrm{Lie} \, H$ that is nonzero and lifts to a Lie p -subalgebra $\mathfrak{h}'_1$ of $\mathrm{Lie} \, E$. By *loc. cit.*, there is a subgroup H_1 of H such that $\mathrm{Lie} \, H_1 = \mathfrak{h}_1$, and a lifting H'_1 of H_1 into E such that $\mathrm{Lie} \, H'_1 = \mathfrak{h}'_1$. Applying again the reduction described above, one reduces to the same problem, where H has been replaced by H/H_1 . Since H has height 1, $\mathrm{Lie}(H/H_1) = \mathrm{Lie} \, H / \mathrm{Lie} \, H_1$ (*loc. cit.*), hence $\mathrm{rk} \, \mathrm{Lie}(H/H_1) \leq \mathrm{rk} \, \mathrm{Lie}(H) - 1$. In brief, proceeding by induction on the rank of $\mathrm{Lie} \, H$, one sees that it suffices, when $\mathfrak{h} \neq 0$, to find a nonzero Lie subalgebra of $\mathrm{Lie} \, H$ that lifts into $\mathrm{Lie} \, E$. Now one has the following lemma:

Lemma 5.8.2. *Let k be a field of characteristic $p > 0$, and φ a surjective morphism of Lie p - k -algebras of finite rank $\mathfrak{g} \rightarrow \mathfrak{h}$. Then:*

- i) *If $\mathfrak{h}_{\bar{k}}$ is reductive (4.2.2) and $\neq 0$, there is a reductive Lie subalgebra $\mathfrak{h}_1$ of $\mathfrak{g}$ whose image in $\mathfrak{h}$ is nonzero.*
- ii) *If k is perfect and if u is a unipotent element of $\mathfrak{h}$ (i.e. there is n such that $u^{(p^n)} = 0$), then u lifts to a unipotent element of $\mathfrak{g}$.*

Take for X an element $\neq 0$ of $\mathfrak{h}$ in case i) and u in case ii), and let X' be a lifting of X in $\mathfrak{g}$. The Lie p -subalgebra of $\mathfrak{g}$ generated by X' is an abelian Lie p -algebra (Exp. VII) $\mathfrak{h}'$. *Case i).* It is clear from the description given in 4.2.2 that the reductive part (*loc. cit.*) of $\mathfrak{h}'_{\bar{k}}$ is already defined over k ; denote it by $\mathfrak{r}'$. I say that the image of $\mathfrak{r}'$ in $\mathfrak{h}$ is nonzero. To establish

this point, we can assume that k is algebraically closed, so that $\mathfrak{h}' = \mathfrak{r}' \oplus \mathfrak{u}'$ ($\mathfrak{u}'$ the unipotent part of $\mathfrak{h}'$). If the image of $\mathfrak{r}'$ in $\mathfrak{h}$ were zero, the image of $\mathfrak{h}'$ in $\mathfrak{h}$ would be unipotent, hence would be zero since $\mathfrak{h}$ is reductive (cf. 2.4 ii)), whereas by construction it contains X .

Case ii). One proceeds in the same way, interchanging the roles of $\mathfrak{r}'$ and $\mathfrak{u}'$.

End of the proof of 5.8.1. Assuming $H \neq 0$, there is, by 5.8.2, a nonzero reductive Lie subalgebra $\mathfrak{h}'_1$ of $\text{Lie } E$. Since $\text{Lie } U$ is unipotent (4.3 i)), one necessarily has $(\text{Lie } U) \cap \mathfrak{h}'_1 = 0$, so that $\mathfrak{h}'_1$ is a lifting of a Lie p -subalgebra of $\text{Lie } H$.

End of the proof of 5.1.1 i d). By 5.8.1, there is a family of algebraic subgroups H'_n ($n \in \mathbf{N}$) of E , such that H'_{n+1} contains H'_n and such that H'_n lifts ${}_{F^n}H$. The decreasing sequence of subgroups $\text{Centr}_E(H'_n)$ is stationary; let C be its stationary value. The center Z of C contains H'_n for every n , hence the image of Z in H contains ${}_{F^n}H$ for every n , and consequently is an open subgroup of H (Exp. VIIA §4), hence is equal to H since H is connected. To prove that E is trivial, one can therefore replace E by Z , hence assume that E is commutative. We shall then see in 7.2.1 that E contains a maximal subgroup of multiplicative type M whose formation commutes with extension of the base field. Since $E_{\bar{k}}$ is a trivial extension (5.1.1 i) a)) and U is unipotent, it is clear that M is the unique lifting of H into E .

Proof of 5.1.1 i) c) (U k -solvable). Since $(H/H^0)(k)$ is of order prime to p , it is immediate by duality that there is an integer n such that ${}_nH$ is an étale subgroup and such that the canonical morphism ${}_nH \rightarrow H/H^0$ is an epimorphism, so that $H/{}_nH$ is connected. By 5.2.3 d), there is a lifting H' of ${}_nH$ into E . One shows, as at the beginning of the proof of 5.8.1, that $C = \text{Centr}_E(H')$ is a subgroup of E such that $C \rightarrow H$ is an epimorphism and such that $C \cap U$ is smooth. Replacing E by C and passing to the quotient by H' , one is reduced to the case where U is smooth and H connected, that is, to case i) d).

This completes the proof of 5.1.1.

5.9. Counterexamples. —

Let us first indicate a procedure for obtaining nontrivial extensions of a k -group of multiplicative type H by a unipotent k -group U . Suppose that an action of H on U is given, and let E be the semidirect product $E = U \cdot H$. Let on the other hand an element of $H^1(k, U)$ be represented by a 1-cocycle a . The group U acts by inner automorphisms on E . The datum of a therefore defines a k -form of E , denoted by E_a . Suppose moreover that U is commutative; then U acts trivially on U and on the quotient $E/U = H$, so that E_a is again an extension of H by U . Suppose, for simplicity, that H is an étale diagonalizable group, so that the Galois group $\mathcal{G}$ of $\bar{k}/k$ acts trivially on $H(k)$; the action of $\mathcal{G}$ on the points of $E_a(\bar{k})$ is then given by the formula:

$${}^g(u, h) = ({}^gu + a(g) - {}^ha(g), h) \quad (g \in \mathcal{G}, u \in U(\bar{k}), h \in H(\bar{k})).$$

If h is a point of $H(k)$, and X its inverse image in E_a , X is therefore a torsor under U defined by the class of the 1-cocycle of $\mathcal{G}$ with values in U , $g \mapsto a(g) - {}^ha(g)$. It follows that if there is a point h of $H(k)$ such that the preceding 1-cocycle is nontrivial, the extension E_a is nontrivial. We shall apply this construction in two particular cases:

a) *Nontrivial extension of an étale diagonalizable group H by $\mathbf{Z}/p\mathbf{Z}$.*

Take for H the group $(\mathbf{Z}/p\mathbf{Z})^\times$, which one makes act by multiplication on $U = \mathbf{Z}/p\mathbf{Z}$. Let on the other hand k be a field of characteristic p , K an extension of k with Galois group $\mathcal{G}$ isomorphic to $\mathbf{Z}/p\mathbf{Z}$, and a a nonzero element of $H^1(\mathcal{G}, U) = \text{Hom}(\mathcal{G}, U)$. The group E_a then answers the question.

b) *Example of a nontrivial extension of an étale diagonalizable group H by a smooth connected unipotent group U .*

Take for the field k a nonperfect field such that there exists a k -form U of $\mathbf{G}_a$ such that $H^1(k, U) \neq 0$. For example (cf. J.-P. Serre, *Cohomologie Galoisienne*), one can take for k the fraction field of a discrete valuation ring of equal characteristic $p > 0$, and for U the algebraic subgroup of $\mathbf{G}_a \times_k \mathbf{G}_a$ with equation $X^p + X + tY^p = 0$, where t denotes a uniformizer of k . Indeed, assuming for simplicity that k contains the $(p-1)$ th roots of unity, one has an exact

sequence of algebraic groups:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbf{Z}/p\mathbf{Z} & \longrightarrow & U & \longrightarrow & \mathbf{G}_a \longrightarrow 0 \\ & & & & (x, y) & \longmapsto & y, \end{array}$$

hence an exact cohomology sequence:

$$\mathbf{G}_a(k) \xrightarrow{d} H^1(k, \mathbf{Z}/p\mathbf{Z}) \longrightarrow H^1(k, U) \longrightarrow 0,$$

where d associates to x the homogeneous space under $\mathbf{Z}/p\mathbf{Z}$ with equation

$$X^p + X + tx^p = 0.$$

Moreover, one knows that $H^1(k, \mathbf{Z}/p\mathbf{Z})$ is isomorphic to $k/\mathfrak{P}(k)$ (where $\mathfrak{P}(x) = x^p + x$), hence $H^1(k, U)$ is isomorphic to $k/(\mathfrak{P}(k) + tk^p)$. Suppose moreover $p \neq 2$; it is then clear that t^2 is an element of k that does not belong to $\mathfrak{P}(k) + tk^p$, hence $H^1(k, U) \neq 0$. On the other hand μ_{p-1} acts on U by the formula:

$$(h, x, y) \longmapsto (hx, hy).$$

Denote by $\mathcal{G}$ the Galois group of the extension K defined by the equation

$$X^p + X + t^2 = 0,$$

and let $a \in H^1(\mathcal{G}, U)$ be the nonzero element described above. One immediately checks that E_a is then a nontrivial extension of μ_{p-1} by U .

c) *Nontrivial extension of $\mathbf{G}_m$ by α_p .*

By 5.1.1 i) b), such an extension can exist only over a nonperfect field k . Let therefore k be a nonperfect field, G the semidirect product of $U = \mathbf{G}_a$ by $H = \mathbf{G}_m$, with $\mathbf{G}_m$ acting on $\mathbf{G}_a$ by homotheties. Since U is invariant in G , ${}_F U$ is also. Let $G' = G/{}_F U$. The group G' is isomorphic to $\mathbf{G}_a \cdot \mathbf{G}_m$, where $\mathbf{G}_m$ acts on $\mathbf{G}_a$ by the formula:

$$(h, u) \longmapsto h^p u.$$

The functor $\mathcal{T}_G$ (resp. $\mathcal{T}_{G''}$) of subtori of G (resp. G'') (cf. Exp. XV) is isomorphic to $\mathbf{G}_a$, and the morphism $\mathcal{T}_G \longrightarrow \mathcal{T}_{G''}$ deduced from the morphism $G \longrightarrow G''$ is identified with the morphism $u \longmapsto u^p$. It follows that if T'' is a subtorus of G'' corresponding to a point x of $k \simeq \mathbf{G}_a(k)$ such that $x^{1/p}$ is not in k , the inverse image E of T'' in G will be an extension of a torus T'' by ${}_F U = \alpha_p$, will have no maximal tori defined over k , hence will be nontrivial. One finds for E the subgroup of $G = \text{Spec}[U, T, T^{-1}]$ with equation $U^p = x - xT^p$.

Remark 5.9.1. This last example shows that a nonsmooth algebraic group defined over a nonperfect field does not necessarily have maximal tori, and thus answers the question asked in Exp. XIV 1.5 b).

d) *Let us now give an example of a trivial extension E of a group of multiplicative type H by a unipotent group U , and of two liftings H' and H'' of H that are not conjugate by an element of $U(k)$.*

Take for E the semidirect product of $U = \mathbf{G}_a$ by $\mu_p = \text{Spec } k[T]/(T^p - 1)$, the action of μ_p on $\mathbf{G}_a = \text{Spec } k[U]$ being defined by the comorphism:

$$U \longmapsto (U + U^p)T - U^p.$$

The centralizer of μ_p in U is then the étale group Z with equation $U + U^p = 0$. It follows that if k is not algebraically closed, the canonical map $U(k) \longrightarrow (U/Z)(k)$ is not in general surjective, hence, in view of 5.1.1 ii) b), two liftings of μ_p into E are not necessarily conjugate by an element of $U(k)$.

Here is another example, with k algebraically closed of characteristic $p > 0$. Let G be the radicial group that is the semidirect product of μ_p by α_p , where μ_p acts on α_p by “homotheties”. One then has an exact sequence of algebraic groups with operator group μ_p :

$$\begin{array}{ccccccc} 0 & \longrightarrow & \alpha_p & \longrightarrow & \mathbf{G}_a & \longrightarrow & \mathbf{G}_a \longrightarrow 0 \\ & & & & x & \longmapsto & x^p, \end{array}$$

where μ_p acts by homotheties on the first term $\mathbf{G}_a$, and trivially on the second. The exact cohomology sequence (App. I, prop. 11) gives here the exact sequence:

$$0 \longrightarrow \mathbf{G}_a(k) \longrightarrow H^1(\mu_p, \alpha_p) \longrightarrow H^1(\mu_p, \mathbf{G}_a).$$

Since the last term is zero (I 5.3.3), one sees that $H^1(\mu_p, \alpha_p)$ is nonzero, hence two liftings of μ_p into G are not necessarily conjugate.

6. Extension of a unipotent group by a group of multiplicative type

6.1. Statement of the theorem. —

Theorem 6.1.1. *Let k be a field, U a unipotent algebraic k -group, H a k -group of multiplicative type, and E an algebraic k -group that is an extension of U by H , so that one has the exact sequence*

$$1 \longrightarrow H \longrightarrow E \longrightarrow U \longrightarrow 1.$$

Then the extension E is trivial and there is a unique lifting of U into E in each of the following cases:

- A) *The group U is smooth and one of the following conditions is satisfied:*
 - i) *U is connected and the canonical morphism $E \longrightarrow U$ has a section.*
 - ii) *U has a composition series with successive quotients isomorphic to $\mathbf{G}_a$.*
 - iii) *H is étale.*
 - iv) *k is perfect.*
- B) *$U = \alpha_p$ and k is perfect.*
- C) *E is commutative and k is perfect.*

6.2. Proof of 6.1.1 A). —

Let us first establish three lemmas.

Lemma 6.2.1. *Let S be a prescheme, E a group S -prescheme that is an extension of a group S -prescheme U with connected fibers by an S -group of multiplicative type and of finite type H (i.e. U is the quotient of E by H for the fpqc topology). Then E is a central extension.*

Indeed, since H is commutative, the group U acts by inner automorphisms on H , through a group S -morphism

$$u : U \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr}}(H).$$

The functor $\underline{\mathrm{Aut}}_{S\text{-gr}}(H)$ is representable by an étale S -scheme (Exp. X 5.10), and consequently the identity section is an immersion that is both open and closed. Since U has connected fibers, one deduces that u is the unit morphism.

Lemma 6.2.2. *With the notation of 6.1.1, if E is trivial, there is a unique lifting of U into E .*

Let therefore U' and U'' be two liftings of U into E . To show that $U' = U''$, we can assume that k is algebraically closed, and it suffices to show that $H^1(U, H) = 0$. If U is connected, U centralizes H (6.2.1), hence

$$H^1(U, H) = \mathrm{Hom}_{k\text{-gr}}(U, H) = 0$$

by 2.4 ii). In the general case, denote by U'_1 the unique lifting into E of the connected component U^0 of U , and let $N = \underline{\mathrm{Norm}}_E(U'_1)$. If $g \in E(k)$, $\mathrm{int}(g)U'_1$ is a lifting of U^0 , hence is equal to U'_1 , and consequently $N(k) \supset E(k)$. Moreover, N contains H (6.2.1) and U'_1 , whence immediately the fact that $N = E$. Passing to the quotient by U'_1 , one is reduced to the case where U is étale. In this case, since k is algebraically closed, $H^1(U, H)$ is identified

with the ordinary cohomology group $H^1(U(k), H(k))^{(10)}$ and consequently is zero, since $U(k)$ is a finite p -group and $H(k)$ is uniquely p -divisible.

Corollary 6.2.3. *To prove that E is trivial, it suffices to show that E becomes trivial after finite separable extension of the base field; in particular, one can assume that H is diagonalizable.*

Lemma 6.2.4. *For every central extension E of U by a diagonalizable k -group H to be trivial, it suffices that every central extension of U by $\mathbf{G}_m$ be trivial.*

Indeed, by induction on r , one first notes that the hypothesis made implies that E is trivial if $H = \mathbf{G}_m^r$. In the general case, H embeds into $\mathbf{G}_m^r$ for a suitable integer r (this is immediate by duality); let $H'' = \mathbf{G}_m^r/H$. One obtains the exact sequence (App. I 2.1):

$$Z^1(U, H'') \longrightarrow \text{Ext}_{\text{alg}}(U, H) \longrightarrow \text{Ext}_{\text{alg}}(U, \mathbf{G}_m^r) = 0$$

(where U acts trivially on $H, H'', \mathbf{G}_m^r$). But $Z^1(U, H'') = \text{Hom}_{k\text{-gr}}(U, H'') = 0$ (2.4 ii)), hence $\text{Ext}_{\text{alg}}(U, H) = 0$.

Proof of 6.1.1 A) i). Since $E \rightarrow U$ has a section, E defines an element $\bar{f}$ of $H^2(U, H)$ (App. I 3.1). We must show that $\bar{f}$ is zero, and for this it suffices to show that a 2-cocycle $f : U \times_k U \rightarrow H$ is a constant morphism, which will follow from the following lemma:

Lemma 6.2.5. *Let U be a smooth connected unipotent algebraic k -group, and H a k -group of multiplicative type; then every k -morphism (of preschemes):*

$$f : U \longrightarrow H$$

is constant.

To prove this lemma, we can assume that k is algebraically closed. We proceed by increasing induction on $\dim U$. If $\dim U > 0$, U has a composition series (cf. 3.9):

$$1 \longrightarrow U' \longrightarrow U \xrightarrow{\pi} U'' \longrightarrow 1$$

with $U' \simeq \mathbf{G}_a$ and $\dim U'' < \dim U$. It suffices to show that f factors through U'' . Since the graph of the equivalence relation defined by π is smooth over k , hence reduced, it suffices to show that if $x, y \in U(k)$ have the same image z in $U''(k)$, then $f(x) = f(y)$. Now $\pi^{-1}(z)$ is isomorphic to $\mathbf{G}_a$, hence the restriction of f to $\pi^{-1}(z)$ factors through a reduced irreducible component of H , hence through a k -scheme isomorphic to $(\mathbf{G}_m)^r$. It then suffices to note that every morphism from $\mathbf{G}_a$ to $(\mathbf{G}_m)^r$ is constant, since every invertible regular function on $\mathbf{G}_a$ is constant.

Proof of 6.1.1 A) ii). By 6.2.1, 6.2.3 and 6.2.4, we can assume that $H = \mathbf{G}_m$. By hypothesis, U has a composition series $U \supset U_1 \supset \cdots$ such that U_i/U_{i+1} is isomorphic to $\mathbf{G}_a$. Let E_1 be the inverse image of U_1 in E . By induction on $\dim U$, one can assume that the extension E_1 is trivial; let U'_1 be the unique lifting of U_1 . Proceeding as in the proof of 6.2.4, one shows that U'_1 is invariant in E . After passing to the quotient by U'_1 , one reduces to the case where $U = \mathbf{G}_a$. The S -scheme E (where $S = U$) is then a torsor under the S -group $\mathbf{G}_m \times_k S$, hence has a section, since $\text{Pic}(S) = 0$ (Exp. VIII 4.3). The extension E is then trivial by A) i).

Proof of 6.1.1 A) iii) (U smooth, H étale). Suppose first that U is connected. The group $U_{\bar{k}}$ then has a composition series with successive quotients isomorphic to $\mathbf{G}_a$, so $E_{\bar{k}}$ is trivial by A) ii). Since H is étale, it is clear that the unique lifting of $U_{\bar{k}}$ in $E_{\bar{k}}$ is the connected component of $E_{\bar{k}}$, so this lifting is already defined over k . In the general case, after passing to the quotient by the connected component of E , one reduces to the case where E is étale, then to the case where E is completely split (6.2.3). Since $U(k)$ is a p -group and $H(k)$ has order prime to p , one may take as lifting of $U(k)$ the Sylow p -subgroup of $E(k)$.

⁽¹⁰⁾**N.D.E.** See, for example, Proposition III.6.4.2 of the book by M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson & North-Holland (1970).

Proof of 6.1.1 A) iv) (U smooth, k perfect). If U is connected, U has a composition series with successive quotients isomorphic to $\mathbf{G}_a$ (4.1.2 b)), and one applies A) ii). In the general case, the foregoing allows us to reduce to the case where U is étale, then to the case where U is completely split and H diagonalizable (6.2.3). Using now a characteristic composition series of H ($H \supset H^0 \supset {}_{F^n}(H) \supset 0$), one reduces to the case where H is of one of the following three types:

- a) H is étale.
- b) $H = (\mathbf{G}_m)^r$.
- c) H is radicial.

In case a), one applies A) iii); in case c), one applies 1.6. Finally, in case b), one remarks that by Hilbert's Theorem 90, $E \rightarrow U$ has a section, so that it suffices to show that $H^2(U, (\mathbf{G}_m)^r) = H^2(U(k), (\mathbf{G}_m)^r(k)) = 0$. Now $U(k)$ is a finite p -group, whereas $(\mathbf{G}_m)^r(k)$ is uniquely p -divisible (since k is perfect).

6.3. — Proof of 6.1.1 B) and C). By 6.2.1, 6.2.3 and 6.2.4, one sees that it suffices to prove B) when $H = \mathbf{G}_m$. One therefore has an exact sequence

$$1 \longrightarrow \mathbf{G}_m \longrightarrow E \longrightarrow \alpha_p \longrightarrow 1.$$

Since $\mathbf{G}_m$ is smooth, one deduces an exact sequence of Lie p -algebras (App. II 3.2):

$$(*) \quad 0 \longrightarrow \text{Lie } \mathbf{G}_m \longrightarrow \text{Lie } E \longrightarrow \text{Lie } \alpha_p \longrightarrow 0.$$

The group α_p being of height 1, the extension E is trivial if and only if (App. II 2.2) the exact sequence (*) of Lie p -algebras is split. One knows that $\text{Lie } \alpha_p$ is generated by an element $X \neq 0$ such that $X^{(p)} = 0$ (App. II 2.1); it therefore suffices to show that X lifts to an element Z of $\text{Lie } E$ such that $Z^{(p)} = 0$. Now by 5.8.2 ii), there exists a unipotent element Z of $\text{Lie } E$ lifting X . Since the unipotent part of $\text{Lie } E$ clearly has dimension at most 1, one necessarily has $Z^{(p)} = 0$.

Proof of 6.1.1 C). If U'_1 is a lifting in E of an algebraic subgroup U_1 of U , U'_1 is invariant in E , since E is supposed commutative, and we may pass to the quotient by U'_1 . Using a composition series of U (3.5 ii)), the preceding remark allows us to reduce to the case where U is smooth or equal to α_p . But then E is trivial by A) iv) and B).

6.4. — Examples of extensions of a unipotent group U by a group of multiplicative type H which are not trivial. In view of 6.1.1 A) iv), the problem arises only in characteristic $p > 0$.

- a) $H = \mathbf{G}_m$, U is a nontrivial form of $\mathbf{G}_a$ (cf. App. III, § 5).

Let k be an imperfect field of characteristic 2, u an element of k such that $u^{1/2}$ does not belong to k . Consider the affine group E with ring $k[X, Y, (X^2 + uY^2)^{-1}]$, where multiplication is given by the comorphism

$$(X, Y) \longmapsto (XX' + uYY', XY' + YX').$$

The group E is smooth, connected, commutative, of dimension 2; the subscheme $Y = 0$ defines a subgroup $H \simeq \mathbf{G}_m$. The kernel K of raising to the second power in E has equation

$$X^2 + uY^2 = 1,$$

hence has dimension 1. The group K contains the unipotent radical of E (defined over $\bar{k}$), but also the contribution of H , which is isomorphic to μ_2 . Since K is reduced over k , the unipotent radical of E is not defined over k , and $U = E/H$ does not lift in E . (One immediately checks that U is the form of $\mathbf{G}_a$ with ring $k[V, W]/(V + uV^2 + W^2)$, the morphism $E \rightarrow U$ corresponding to the comorphism $V \mapsto Y^2/(X^2 + Y^2u)$, $W \mapsto XY/(X^2 + Y^2u)$.)

- b) $H = \mathbf{G}_m$, $U = \mathbf{Z}/2\mathbf{Z}$, k imperfect of characteristic 2.

Choosing k and u as in a), consider the subgroup E of GL_2 generated by the element X such that

$$X = \begin{pmatrix} 0 & 1 \\ u & 0 \end{pmatrix} = \begin{pmatrix} u^{1/2} & 0 \\ 0 & u^{1/2} \end{pmatrix} \begin{pmatrix} 0 & u^{-1/2} \\ u^{1/2} & 0 \end{pmatrix}.$$

The group E is an extension of $\mathbf{Z}/2\mathbf{Z}$ by $\mathbf{G}_m$, but this extension is not trivial, since the unipotent part of X is not defined over k .

c) $H = \mu_p$, $U = \alpha_p$, k imperfect.

Let $\mathfrak{e}$ be the commutative Lie p -algebra generated by two elements X and Y such that $X^{(p)} = X$ and $Y^{(p)} = aX$. By App. II 2.2, $\mathfrak{e}$ is the Lie p -algebra of an algebraic group E , an extension of α_p by μ_p , but this extension is trivial if and only if there exists $b \in k$ such that $b^p = a$ (for one then has $(bX + Y)^{(p)} = 0$).

d) $H = \mu_2$, $U = \alpha_2 \times \alpha_2$, E noncommutative, k a field of characteristic 2.

Consider the special linear group $\mathrm{SL}_{2,k}$ and let $E = {}_F(\mathrm{SL}_{2,k})$. The group E is a radicial group of height 1, whose Lie algebra is generated by three elements X, Y, Z satisfying the following relations:

$$[X, Y] = Z, \quad [X, Z] = [Y, Z] = 0,$$

$$X^{(p)} = Y^{(p)} = 0, \quad Z^{(p)} = Z.$$

Consequently, E is a central extension of $U \simeq \alpha_2 \times \alpha_2$ by μ_2 . Each factor α_2 of U lifts uniquely in E , but U itself does not lift in E , since $[X, Y] \neq 0$.

7. Nilpotent affine algebraic groups

7.1. — Extensions of groups of multiplicative type.

Proposition 7.1.1. *Let S be a prescheme, H and K two group S -preschemes of multiplicative type and of finite type, E a group prescheme which is an extension of K by H (i.e. K is the quotient of E by H for the fpqc topology). Then E is of multiplicative type in the following two cases:*

- a) E is commutative.
- b) The fibers of K are connected.

Proof. i) Case where S is the spectrum of a field k . The assertion to be proved is local for the fpqc topology, which allows us to suppose k algebraically closed, hence H and K diagonalizable. Note that K acts trivially on H by inner automorphisms: this is clear in case a), and follows from 6.2.1 in case b). By induction on the length of a suitable composition series of K , one reduces to the case where K is of one of the following three types: a) $K = \mathbf{G}_m$, b) $K = \mu_p$, c) $K = \mu_q$ with $(q, p) = 1$ and, in this case, E is commutative. Using now an embedding of H in $(\mathbf{G}_m)^r$, one deduces that E embeds in an extension of K by $(\mathbf{G}_m)^r$. One may therefore suppose $H = (\mathbf{G}_m)^r$.

a) If $K = \mathbf{G}_m$, E is a smooth connected affine algebraic group (Exp. VIB 9.2) of unipotent rank zero, hence is a torus.

b) $K = \mu_p$. In this case, E is a trivial extension. Indeed, since H is smooth, the canonical morphism $\mathrm{Lie} E \rightarrow \mathrm{Lie} \mu_p$ is surjective (App. II 3.2), and it suffices to apply 5.8.2 i), taking account of App. II 2.2.

c) $K = \mu_q$, with $(q, p) = 1$ and E commutative. Here again, the extension E is trivial. Indeed, let $x \in E(k)$ be a lifting of a generator $\bar{x}$ of $\mu_q(k)$. The element x^q is an element of $H(k)$, hence is of the form y^q , $y \in H(k)$ (note that $(\mathbf{G}_m)^r(k)$ is q -divisible). Since E is commutative, $y^{-1}x$ is a lifting of $\bar{x}$ of order q .

ii) General case. The groups H and K are flat, affine and of finite presentation over S (Exp. IX 2.1), and consequently the same holds for E (Exp. VIB 9.2). Using then the general technique of VIB § 10, we reduce to the case where S is noetherian. To show that E is of

multiplicative type, it then suffices to prove that E is commutative and that ${}_nE$ is finite over S for every n (Exp. X 4.8 b)).

a) E is commutative. One must verify that the morphism

$$E \times_S E \longrightarrow E, \quad (x, y) \longmapsto [x, y] = xyx^{-1}y^{-1}$$

factors through the unit section of E , and it suffices to verify this when S is the spectrum of an artinian local ring. But then E is of multiplicative type by i) and Exp. X 2.3.

b) ${}_nE$ is finite over S . Indeed, one has the exact sequence

$$0 \longrightarrow {}_nH \longrightarrow {}_nE \longrightarrow {}_nK \xrightarrow{u} H_n$$

(where H_n is the cokernel of raising to the n th power in H). One knows that H_n is of multiplicative type (Exp. IX 2.7), hence separated, and that ${}_nH$ and ${}_nK$ are finite over S ; $\text{Ker } u$ is a closed subgroup of ${}_nK$, hence is finite over S , and ${}_nE$ is finite over S as an extension of a finite group by a finite group (Exp. VIB 9.2).

7.2. — Structure of commutative affine algebraic groups.

Theorem 7.2.1. *Let k be a field, G a commutative affine algebraic k -group. Then:*

- a) *G contains a largest subgroup of multiplicative type M . The group M is characteristic in G and G/M is unipotent, and its formation commutes with extension of the field k .*
- b) *If k is perfect, G is the direct product of M and a unipotent algebraic subgroup U , and this uniquely.*

Proof. i) k algebraically closed. When G is smooth, 7.2.1 b) is well known (BIBLE § 4 Th. 4). If G is radicial of height 1, the decomposition of $\text{Lie } G$ described in 4.2.2 corresponds, taking account of 4.3.1 v) and App. II 2.2, to a decomposition of G of the type 7.2.1 b). In the general case, G admits a composition series whose successive quotients are smooth or radicial of height 1 (App. II 3.1). To prove 7.2.1 b), it then suffices to note that if one has an exact sequence of commutative algebraic groups

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0,$$

where G' (resp. G'') is the product of a group of multiplicative type and a unipotent group $G' = M' \cdot U'$ (resp. $G'' = M'' \cdot U''$), then the same holds for G . Indeed, consider the exact sequence

$$0 \longrightarrow G'/U' \longrightarrow G/U' \longrightarrow G'' \longrightarrow 0.$$

By 7.1.1 a), the inverse image in G/U' of M'' is a subgroup of multiplicative type M_1 . The group M_1 lifts to a subgroup M of G (5.1.1 i) a)). Similarly, using this time 6.1.1 C), one proves that there exists a unipotent subgroup U of G which is an extension of U'' by U' . It is clear that $G = M \cdot U$.

ii) k arbitrary. By i), $G_{\bar{k}}$ is the direct product of a group of multiplicative type $M_{\bar{k}}$ with a unipotent group $U_{\bar{k}}$. Put $S = \bar{k} \times_k \bar{k}$, and let M_1 and M_2 be the two inverse images of $M_{\bar{k}}$ by the two projections $S \rightrightarrows \bar{k}$. The group $G_S/M_2 = (G/M_{\bar{k}})_S$ has unipotent fibers, so the image of M_1 in G_S/M_2 is zero (2.4 i)), and M_2 contains M_1 . Similarly, M_1 contains M_2 , and finally $M_1 = M_2$. By fpqc descent, it follows that $M_{\bar{k}}$ comes from an algebraic subgroup M of G . It is clear that M is of multiplicative type, that G/M is unipotent, and that formation of M is compatible with every extension of the field k . For every k -prescheme S , every subgroup of multiplicative type H of G_S is contained in M_S . Indeed, by 2.5 its image in the group with unipotent fibers $(G/M)_S = G_S/M_S$ is zero. Taking in particular for H the transform of M_S by an automorphism of G_S , one deduces that M is characteristic in G .

Finally, if k is perfect, G/M lifts in G to a unipotent group, and this uniquely by 6.1.1 C).

Remark 7.2.2. i) If k is imperfect, the unipotent component of $G_{\bar{k}}$ is not necessarily defined over k , as shown by example 6.4 a).

- ii) Contrary to what happens for the component of multiplicative type M , the unipotent component U is in general not characteristic in G (and this regardless of the characteristic of k). Of course, uniqueness of the decomposition 7.2.1 b) implies that U is invariant under every k -automorphism of G . But if U and M are such that there exist a k -prescheme S and a nonzero S -homomorphism $h : U_S \rightarrow M_S$ (cf. 2.6), one deduces an S -automorphism of G_S , $(u, m) \mapsto (u, h(u) + m)$, which does not leave U_S invariant.
- iii) If G is finite over k , G/M corresponds by Cartier duality (2.6) to the connected component of the dual $D(G)$ of G .

7.3. — Structure of nilpotent affine algebraic groups.

Theorem 7.3.1. *Let k be a field, G a nilpotent algebraic k -group (Exp. VIB § 8), affine, connected. Then G has a largest subgroup of multiplicative type M . The group M is central and characteristic, and G/M is a unipotent algebraic group.*

Let Z be the center of G , M the largest subgroup of multiplicative type of Z (7.2.1). Since Z is characteristic in G and M characteristic in Z , M is characteristic in G . It suffices to show that G/M is unipotent. By induction on the length of the ascending central series of G , this will follow more generally from the following lemma:

Lemma 7.3.2. (Rosenlicht). *Let G be a connected algebraic k -group, Z its center. Then the center Z' of $G' = G/Z$ is unipotent.*

Proof. We may suppose k algebraically closed. It then suffices to show that Z' contains no subgroup isomorphic to μ_ℓ for any prime number ℓ (4.6.1 vi)). Let therefore μ_ℓ be a subgroup of Z' , N its inverse image in G . Since μ_ℓ is central in G' , N is invariant in G .

i) *Case where $(\ell, p) = 1$.* One can find an element x of $N(k)$ and an integer n having the following properties:

- a) x lifts a generator $\bar{x}$ of μ_ℓ .
- b) $x^{\ell^n} \in Z^0(k)$;

(it suffices to choose a lifting of $\bar{x}$ whose image in $N/Z^0(k)$ belongs to the Sylow ℓ -subgroup).

Raising to the ℓ th power in the commutative group Z^0 is an étale morphism, so $Z^0(k)$ is ℓ -divisible. Consequently, after multiplying x by an element of $Z^0(k)$, one may suppose that $x^{\ell^n} = 0$. The group N is then generated by two commutative groups which commute (Z and the group generated by x), hence is commutative. The group ${}_{\ell^n}N$ is a group of multiplicative type, characteristic in N , hence invariant in G , and consequently central, G being connected (Exp. IX 5.5). Thus ${}_{\ell^n}N$ is contained in Z , which contradicts the fact that its image in G' contains μ_ℓ .

ii) $\ell = p$. There then exists an integer n such that the image of ${}_{F^n}N$ in G' contains μ_p , so that one has the exact sequence

$$(*) \quad 1 \longrightarrow K \longrightarrow {}_{F^n}N \longrightarrow \mu_p \longrightarrow 1.$$

The group K is contained in Z , hence is commutative; it then follows from 7.2.1 and 7.1.1 b) and 5.5.1 that there exists a subgroup of multiplicative type contained in ${}_{F^n}N$ whose image in the quotient by K is μ_p . One deduces as in i) that ${}_{F^n}N$ is commutative. The component of multiplicative type of ${}_{F^n}N$ (7.2.1) is characteristic in ${}_{F^n}N$, hence invariant in G , hence central (Exp. IX 5.5); since its image in G' contains μ_p , one obtains a contradiction.

A. Appendix I. Hochschild cohomology and extensions of algebraic groups

A.1. — Definition of the cohomology groups. Let k be a field, G an algebraic k -group, $(\text{Ab})_G$ the abelian category of commutative algebraic k -groups on which G acts. If $A \in \text{Ob}(\text{Ab})_G$, the functor $h_A : ((\text{Sch})/k)^\circ \rightarrow (\text{Ens})$ canonically defined by A is a G - $\mathbf{Z}$ -module in the sense of I 3.2. We may therefore consider the standard complex $C^\bullet(G, A)$

of algebraic cochains of G with values in A (Exp. I 5.1), as well as the group of i -cocycles $Z^i(G, A)$, of i -coboundaries $B^i(G, A)$ and the i th cohomology group $H^i(G, A)$. As usual, $H^0(G, A)$ is identified with the group $A^G(k)$, where A^G is the k -functor of invariants of A under G . The group $H^1(G, A)$ classifies trivial principal homogeneous spaces under A on which G acts.

The functor $A \mapsto H^\bullet(G, A)$ is in general not a cohomological functor from the category $(\text{Ab})_G$ with values in (Ab) ; however, one has the following proposition:

Proposition A.1.1. *Let $0 \rightarrow A' \xrightarrow{u} A \xrightarrow{v} A'' \rightarrow 0$ be an exact sequence in $(\text{Ab})_G$. Then:*
a) If v has a section (that is to say, if there exists a k -morphism of preschemes $s : A'' \rightarrow A$ such that $vs = 1_{A''}$), one has the usual exact cohomology sequence

$$\cdots \rightarrow H^i(G, A) \rightarrow H^i(G, A'') \xrightarrow{d} H^{i+1}(G, A') \rightarrow \cdots$$

b) If $A(k) \rightarrow A''(k)$ is surjective, one has the exact sequence

$$(1) \quad 0 \rightarrow A'^G(k) \rightarrow A^G(k) \rightarrow A''^G(k) \xrightarrow{d} H^1(G, A') \rightarrow H^1(G, A) \rightarrow H^1(G, A'').$$

Proof. a) One notes that existence of a section implies exactness of the sequence of complexes

$$0 \rightarrow C^\bullet(G, A') \rightarrow C^\bullet(G, A) \rightarrow C^\bullet(G, A'') \rightarrow 0.$$

b) If $x'' \in A''^G(k)$, its inverse image in A is a trivial principal homogeneous space under A' (since $A(k) \rightarrow A''(k)$ is supposed surjective) on which G acts, hence defines an element $d(x'') \in H^1(G, A')$. Exactness of sequence (1) is then immediate.

A.2. — The group $\text{Ext}_{\text{alg}}(G, A)$. Let A and G be two algebraic k -groups, E and E' two algebraic k -groups which are extensions of G by A . These two extensions are said to be isomorphic if there exists a group k -morphism $u : E \rightarrow E'$ making the diagram commutative:

$$\begin{array}{ccccc} & & E & & \\ & \nearrow & \downarrow u & \searrow & \\ A & & & & G \\ & \searrow & \downarrow & \nearrow & \\ & & E' & & \end{array}$$

The group E acts on A by inner automorphisms, and if A is commutative, this action factors through G , so G acts on A . Conversely, if $A \in \text{Ob}(\text{Ab})_G$, we denote by $\text{Ext}_{\text{alg}}(G, A)$ the set of classes of algebraic extensions E of G by A for which the action of G on A defined by E and that coming from the structure of an object of $(\text{Ab})_G$ coincide.

$\text{Ext}_{\text{alg}}(G, A)$ is naturally a bifunctor covariant in A and contravariant in G . More precisely:
 a) If $f : A \rightarrow B$ is a morphism in $(\text{Ab})_G$ and E represents an element of $\text{Ext}_{\text{alg}}(G, A)$, one defines $f_*(E) \in \text{Ext}_{\text{alg}}(G, B)$ as the class of the extension of G by B equal to the quotient of the semidirect product $B \cdot E$ (E acting on B through G) by the algebraic subgroup which is the image of A under the morphism (f, i) (this quotient is representable by Exp. VIA § 5), so that one has a commutative diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & A & \xrightarrow{i} & E & \longrightarrow & G \longrightarrow 1 \\ & & \downarrow f & & \downarrow & & \downarrow 1_G \\ 1 & \longrightarrow & B & \longrightarrow & f_*(E) & \longrightarrow & G \longrightarrow 1. \end{array}$$

b) If $g : H \rightarrow G$ is a k -morphism of algebraic k -groups and E is an extension of G by A , the fibered product $E \times_G H$ is naturally an extension of H by A , denoted by $g^*(E)$. One

therefore has a commutative diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & A & \longrightarrow & g^*(E) & \longrightarrow & H \longrightarrow 1 \\ & & \downarrow 1_A & & \downarrow & & \downarrow g \\ 1 & \longrightarrow & A & \longrightarrow & E & \longrightarrow & G \longrightarrow 1. \end{array}$$

Adapting the proofs given in J.-P. Serre, *Groupes algébriques et corps de classe*, chap. VII, one equips $\text{Ext}_{\text{alg}}(G, A)$ with a natural abelian group structure, functorial in A and G .

Proposition A.2.1. *Let $0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0$ be an exact sequence in $(\text{Ab})_G$; then:*
a) *One has a canonical exact sequence of abelian groups*

$$Z^1(G, A) \longrightarrow Z^1(G, A'') \xrightarrow{d} \text{Ext}_{\text{alg}}(G, A') \longrightarrow \text{Ext}_{\text{alg}}(G, A) \longrightarrow \text{Ext}_{\text{alg}}(G, A'').$$

b) *If $A(k) \longrightarrow A''(k)$ is surjective, one deduces from a) the exact sequence*

$$(2) \quad H^1(G, A) \longrightarrow H^1(G, A'') \xrightarrow{d} \text{Ext}_{\text{alg}}(G, A') \longrightarrow \text{Ext}_{\text{alg}}(G, A) \longrightarrow \text{Ext}_{\text{alg}}(G, A'').$$

The exact sequence of a) generalizes the usual exact sequence of the $\text{Hom}(\cdot, \cdot)$ valid in the context of commutative extensions (loc. cit.) and is proved in the same way. Let us simply recall the definition of the coboundary $d : Z^1(G, A'') \longrightarrow \text{Ext}_{\text{alg}}(G, A')$. For this, consider the extension

$$1 \longrightarrow A' \longrightarrow A \cdot G \longrightarrow A'' \cdot G \longrightarrow 1,$$

deduced in an evident way from the exact sequence $0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0$. If $u \in Z^1(G, A'')$, u defines in the usual way a section homomorphism $u : G \longrightarrow A'' \cdot G$. One then has $d(u) = u^*(A \cdot G)$.

A.3. — Comparison of $H^2(G, A)$ and $\text{Ext}_{\text{alg}}(G, A)$. It is well known, in the case of abstract groups, that there exists a functorial isomorphism between the abelian groups $H^2(G, A)$ and $\text{Ext}(G, A)$. Similarly, in the present case, if A is an element of $(\text{Ab})_G$, to every 2-cocycle $u \in Z^2(G, A)$ one can associate an algebraic group structure on the prescheme $A \times_k G$ making it an element of $\text{Ext}_{\text{alg}}(G, A)$. Moreover, this extension is trivial if and only if $u \in B^2(G, A)$ (cf. Exp. III 1.2.2). Recall that the composition law on $A \times G$ is defined by the formula

$$(a, g)(a', g') = a + {}^g a' + u(g, g').$$

It is clear that the extensions of G by A thus obtained are not arbitrary, since they have a section. But conversely, if $E \in \text{Ext}_{\text{alg}}(G, A)$ has a section s , E is isomorphic to the extension of G by A associated to the 2-cocycle u such that

$$u(g, g') = s(g)s(g')s(gg')^{-1}.$$

One finally obtains the following proposition:

Proposition A.3.1. *There exists a functorial isomorphism between the bifunctors with values in abelian groups*

$$(G, A) \longmapsto H^2(G, A) \quad \text{and} \quad (G, A) \longmapsto \text{Ext}_s(G, A),$$

where $\text{Ext}_s(G, A)$ denotes the subgroup of $\text{Ext}_{\text{alg}}(G, A)$ formed by the classes of extensions of G by A which have a section.

B. Appendix II. Recollections and complements on radicial groups

Let p be a prime number > 1 , and let S be an $\mathbf{F}_p$ -prescheme.

B.1. — The Frobenius morphism. For every S -prescheme X and every integer $n > 0$, denote by $P^n : X \longrightarrow X$ the $\mathbf{F}_p$ -endomorphism corresponding to raising to the p^n th power in $\mathcal{O}_X$, and denote by $X^{(n)}$ the S -prescheme which is the inverse image of X under the

morphism $P^n : S \rightarrow S$. There then exists a unique S -morphism $F^n : X \rightarrow X^{(n)}$ making the diagram commutative:

$$\begin{array}{ccc}
 & & X \\
 & \swarrow P^n & \searrow F^n \\
 X & \xleftarrow{\quad} & X^{(n)} \\
 \downarrow & & \downarrow \\
 S & \xleftarrow{P^n} & S
 \end{array}$$

It is clear that F^n is identified with the “ n th iterate” of $F^1 = F$, called the Frobenius morphism of X/S .

If G is a group S -prescheme, $G^{(n)}$ is a group S -prescheme, and $F^n : G \rightarrow G^{(n)}$ is a group S -morphism. Its kernel $_{F^n}(G)$ is a characteristic group subscheme of G (i.e. stable under the functor $\text{Aut}_{S\text{-gr}}(G)$), radical over S . If G is a radical group S -prescheme, one says that G has height $\leq h$ if $_{F^h}(G) = G$.

B.2. — Groups and Lie p -algebras. If G is a group S -prescheme, $\text{Lie}(G)$ (Exp. II) is naturally a restricted Lie p -Algebra (Exp. VIIA §§ 5 and 6). In particular, one has the following result (cf. Exp. VIIA):

Proposition B.2.1. *i) $\text{Lie}(\alpha_p)_S = \text{Lie}(\mathbf{G}_a)_S^{(11)}$ is a sheaf of $\mathcal{O}_S$ -modules, free over $\mathcal{O}_S$ of rank 1, generated by an element X such that $X^{(p)} = 0$.
 ii) $\text{Lie}(\mu_p)_S = \text{Lie}(\mathbf{G}_m)_S$ is a sheaf of $\mathcal{O}_S$ -modules, free over $\mathcal{O}_S$ of rank 1, with a canonical basis X such that $X^{(p)} = X$.*

Let us now recall the fundamental result proved in Exp. VIIA § 7:

Theorem B.2.2. *Suppose S affine with ring A . Then the functor*

$$G \mapsto \text{Lie } G$$

establishes an equivalence of categories between, on the one hand, the category of group S -preschemes G of finite presentation and flat over S , of height 1, whose Lie algebra is locally free over $\mathcal{O}_S$, and, on the other hand, the category of restricted Lie p - A -algebras locally free of finite rank.

Moreover, if G is as above and H is a group S -prescheme of finite presentation, the canonical morphism

$$\text{Hom}_{S\text{-gr}}(G, H) \rightarrow \text{Hom}_{p\text{-}A\text{-Lie}}(\text{Lie } G(S), \text{Lie } H(S))$$

is an isomorphism.

B.3. — Radical groups and smooth groups. We now suppose that S is the spectrum of a field k of characteristic p .

Recall that in Exp. VIA § 5, it was shown that if G is an algebraic k -group and H an algebraic subgroup of G , then the sheaf G/H (a sheaf for the fpqc topology) is representable. Let us then recall (VIIA 8.3):

Proposition B.3.1. *Let G be an algebraic k -group. Then there exists an integer m such that for every $n \geq m$, the algebraic group $G/_{F^n}(G)$ is smooth over k .*

Proposition B.3.2. *Consider an exact sequence of algebraic k -groups*

$$1 \rightarrow G' \rightarrow G \xrightarrow{u} G'' \rightarrow 1$$

⁽¹¹⁾**N.D.E.** The definition of α_p , already given at the beginning of this exposé, has been deleted here.

and the following assertions:

- i) The morphism u is smooth.
- ii) G' is smooth over k .
- iii) For every integer $n > 0$, one has the exact sequence

$$1 \longrightarrow F^n(G') \longrightarrow F^n(G) \longrightarrow F^n(G'') \longrightarrow 1.$$

- iv) The morphism ${}_FG \longrightarrow {}_FG''$ is an epimorphism.
 - v) The morphism $(\operatorname{Lie} G)(k) \longrightarrow (\operatorname{Lie} G'')(k)$ is surjective.
- Then one has the following implications:

$$i) \Longleftrightarrow ii) \implies iii) \implies iv) \Longleftrightarrow (v).$$

Moreover, if G is smooth, the five assertions are equivalent.

- i) $\Leftrightarrow$ ii) by Exp. VIB 9.2 vii).
- ii) $\Rightarrow$ iii). If G' is smooth, $F^n : G' \longrightarrow G'^{(n)}$ is an epimorphism, and iii) follows from the “snake” diagram.
- iii) $\Rightarrow$ iv) is clear.
- iv) $\Leftrightarrow$ v) by Theorem B.2.2.
- v) $\Rightarrow$ ii) when G is smooth. Indeed, G'' is then smooth, and v) implies that one has $\dim G' = \dim_k(\operatorname{Lie} G')(k)$, so G' is smooth.

C. Appendix III. Remarks and complements concerning exposés XV, XVI, XVII

C.1. — It may be that Propositions 1.2 and 1.2 bis of XV remain true if one removes the hypothesis that H_0 is smooth over S_0 . This is in particular the case if G is finite, flat and commutative.

C.2. — Complement to XV 4.8. The following proposition, as well as Theorems C.3.1 and C.4.1 below, will appear in an article in preparation by M. Raynaud on group schemes over a discrete valuation ring.⁽¹²⁾

Proposition C.2.1. *Let S be the spectrum of a discrete valuation ring, t its generic point, G a group S -prescheme of finite type and flat, $\tilde{G} = \operatorname{Spec} \Gamma(G, \mathcal{O}_G)$, $u : G \longrightarrow \tilde{G}$ the canonical morphism. Then:*

- (1) $\tilde{G}$ is naturally a group S -scheme, and u is a homomorphism.
- (2) $\operatorname{Ker}(u)$ is flat over S , and $(\tilde{G}, u)$ is an fpqc quotient of G by $\operatorname{Ker}(u)$, so that $\tilde{G}$ is the largest affine quotient of G .
- (3) If G_t is affine, $\operatorname{Ker}(u)$ is an étale group over S , equal to the unit group if and only if G is separated over S . In particular, a group S -scheme G , flat, of finite type, separated, with affine generic fiber, is affine.

C.3. — In the statement of XV 6.6, the hypothesis that H is equal to its connected normalizer is unnecessary for the set of points s of S such that H_s is a parabolic subgroup of G_s to be an open set. Indeed, let us return to the proof given in paragraph c) preceding Lemma 6.9, and denote by $\bar{N}$ the schematic closure of N_t in G . Thus $\bar{N}$ is a closed group subscheme of G , flat over S , containing H and contained in N . It then follows from the theorem below that $G/\bar{N}$ is representable. Since $G_s/\bar{N}_s$ is proper and connected, one finishes as in loc. cit.

⁽¹²⁾**N.D.E.** Comments to be added here, including references to VIB...

Theorem C.3.1. ⁽¹³⁾ *Let S be the spectrum of a henselian discrete valuation ring, G a group S -prescheme locally of finite type, H a closed group subprescheme of G , flat over S ; then G/H is representable.*

C.4. — Complement to (XVI 1.1). The following theorem makes (VIII 7.9) more precise.

Theorem C.4.1. *Let S be the spectrum of a discrete valuation ring of residue characteristic $p > 0$, and let G be a group S -scheme, commutative, smooth, of finite type and separated over S . Then the following conditions are equivalent:*

- (i) ${}_pG$ is finite over S .
- (ii) *For every S -prescheme S' and every group S' -prescheme H' of finite presentation over S' , separated over S' , every S' -monomorphism $u : G_{S'} \rightarrow H'$ is an immersion.*

C.5. — Example (XVII 6.4 a)) provides an example of a smooth group over a field k whose unipotent radical is not defined over k . The following proposition gives a general method for obtaining such groups:

Proposition C.5.1. *Let k be a field, K a finite radical extension of k of degree > 1 , H a connected smooth algebraic K -group of dimension r , $G = \prod_{K/k} H/K$, which is a smooth connected algebraic k -group of dimension $r[K : k]$. Let $u : G_K \rightarrow H$ be the canonical homomorphism, and $R = \text{Ker}(u)$. Then:*

- i) *The morphism u is an epimorphism, and R is a smooth unipotent connected algebraic group.*
- ii) *If U is a smooth algebraic subgroup of G such that U_K contains R , then $U = G$.*

Corollary. *Keep the notation of the preceding proposition.*

- a) *If H is not unipotent, the unipotent radical of $G_{\bar{k}}$ is not defined over k .*
- b) *If H is a nonzero abelian variety, G is not an extension of an abelian variety by a smooth linear group.*
- c) *If H is not solvable, the solvable radical of $G_{\bar{k}}$ is not defined over k , and G has no Borel subgroup defined over k .*

Let us first show how the corollary follows from the proposition.

- a) Let U be a unipotent radical of G . Then U_K is the unipotent radical of G_K , hence contains R , since R is smooth unipotent connected by i), so $U = G$ by ii). Now G_K admits H as a quotient, and H is not unipotent by hypothesis, so G is not unipotent, whence a contradiction.
- b) If G is an extension of an abelian variety by a linear group L , smooth over k , L_K necessarily contains R , so $L = G$. But G cannot be a linear group, since G_K has a quotient which is a nonzero abelian variety.
- c) Let S (resp. B) be a solvable radical of G (resp. a Borel subgroup of G); then S_K (resp. B_K) contains R , so $S = G$ (resp. $B = G$). But then G is solvable, which contradicts the fact that G_K has a quotient which is not solvable.

Proof of Proposition 5.1: i) Let us begin by describing the morphism u . The datum of a K -scheme

$$f : S \rightarrow \text{Spec}(K)$$

allows one to construct the diagram

$$\begin{array}{ccc} \text{Spec}(K) & \xleftarrow{h} & T \\ \downarrow j & \swarrow f & \downarrow j_T \\ \text{Spec}(k) & \xleftarrow{g} & S \end{array} \quad \begin{array}{c} \uparrow s \\ \downarrow \end{array}$$

⁽¹³⁾**N.D.E.** See Theorem 4C in: S. Anantharaman, *Schémas en groupes, espaces homogènes et espaces algébriques sur une base de dimension 1*, Bull. Soc. Math. France, Mém. 33 (1976), 5–79.

where $g = j \circ f$, $T = \operatorname{Spec}(K) \times_{\operatorname{Spec}(k)} S$, h and j_T are the two projections, and s is the section of T over S such that $h \circ s = f$. The map

$$u(S) : G_K(S) \longrightarrow H(S)$$

is simply the composite map

$$G_K(S) \xrightarrow{\sim} H(T) \longrightarrow H(S),$$

where the last arrow is defined by the section s .

Take in particular for S the spectrum of an algebraic closure $\bar{k}$ of k , and for f the unique k -morphism $K \longrightarrow \bar{k}$, so that T is a local artinian scheme. To prove i), it suffices to do so after the base field extension $K \longrightarrow \bar{k}$. Now it is clear that $G_S = \prod_{T/S} H_T/T$ represents the Greenberg functor of H_T relative to S (M. J. Greenberg, *Schemata over local rings*, Ann. of Maths. 73, 1961, pp. 624–648). The description above then shows that, under this last identification, u_S is the canonical transition morphism

$$\operatorname{Green}(H_T) \longrightarrow H_S = H_T \times_T S.$$

Assertion i) then follows from the fact that H is smooth over K and from (M. J. Greenberg, *Schemata over local rings II*, Ann. of Maths. 78, 1963, pp. 256–266).

ii) To establish ii), we may suppose k separably closed. Let therefore U be a smooth algebraic subgroup of G such that $U_K \supset R$, and let us show that $U = G$. Since G is connected, one may suppose U connected. Let V be the smooth connected algebraic subgroup $u(U_K)$ of H , and $V' = \prod_{K/k} V/K$, which is a smooth connected algebraic subgroup of G . The group V'_K is an algebraic subgroup of G_K , and the canonical morphism $V'_K \longrightarrow V$ is simply the restriction of u to V'_K . By hypothesis, U_K contains R ; a fortiori, U_K contains $\operatorname{Ker}(V'_K \longrightarrow V)$; on the other hand, by construction, the image of U_K in H is equal to V . One deduces that U_K contains V'_K , hence that U contains V' . On the other hand, the canonical isomorphism

$$G(k) \hookrightarrow G(K) \xrightarrow{u(K)} H(K)$$

plainly sends $U(k)$ into $V(K) = V'(k)$; this says that $U(k)$ is contained in $V'(k)$. Since U is smooth and k separably closed, this implies $U \subset V'$, whence $U = V'$. One then has the equalities

$$([K : k] - 1) \dim V = \dim \operatorname{Ker}(V'_K \longrightarrow V) = \dim R = ([K : k] - 1) \dim H.$$

Since $K \neq k$, one concludes that $\dim V = \dim H$, whence $V = H$ and finally $U = V' = G$.